

A Systematic Review of Artificial Intelligence in Piano Education: Efficacy, Ethics, and the Transformation of the Teacher's Role (2020–2025)

LIU CHANG

lc58585533@gmail.com

Handan College <https://orcid.org/0000-0003-3162-2358>

Systematic Review

Keywords: Artificial intelligence, Piano education, Learning efficacy, Ethics, Teacher role transformation

Posted Date: March 8th, 2026

DOI: <https://doi.org/10.21203/rs.3.rs-8998934/v1>

License: © ⓘ This work is licensed under a Creative Commons Attribution 4.0 International License.

[Read Full License](#)

Additional Declarations: The authors declare no competing interests.

Abstract

Technology has begun reshaping piano pedagogy, moving instructional practices away from traditional apprenticeship models toward data-informed learning environments. This systematic review and meta-analysis, conducted according to PRISMA guidelines, examined empirical research on artificial intelligence (AI) in piano education published between 2020 and 2025. Searches of Web of Science, Scopus, and ProQuest identified 391 initial records, of which 62 studies (N = 2,812 participants) satisfied inclusion criteria. Random-effects meta-analysis showed a moderate pooled effect size ($d = 0.442$, 95% CI [0.398, 0.486], $p < .0001$) for AI-supported interventions. Subgroup analyses revealed similar effects across technical skills ($d = 0.462$), musical expression ($d = 0.432$), and learning motivation ($d = 0.434$), with considerable heterogeneity ($I^2 = 61.1\%$) pointing to context-dependent effectiveness. Publication bias assessment indicated no marked asymmetry (Egger's test: $p = .191$). Results show that AI tools correlate with favorable outcomes in psychomotor skill acquisition and, importantly, musical expressiveness—countering assumptions that AI benefits apply only to quantifiable outcomes. Evidence on long-term learning trajectories and higher-order artistic judgment remains sparse. Ethical issues include algorithmic bias, data privacy, and potential homogenization of creative expression. The analysis points to a reconfiguration rather than replacement of the teacher's role, with educators serving as aesthetic mentors and pedagogical decision-makers. This meta-analysis offers the first empirical benchmark for AI-enhanced piano education, underscoring the need for context-sensitive integration, thorough teacher training, and ethically grounded research approaches.

1. Introduction

In recent years, AI has made marked inroads into music education, changing how piano instruction takes place (Cui, 2023; Konovalova et al., 2025; Xiao, 2024). Researchers now test whether large language models can perceive emotional nuance in classical piano playing—a question that seemed improbable just a decade ago (Amin, 2024; Wang, 2025). Studies indicate that AI technology can boost students' practice efficiency and engagement via real-time feedback, intelligent assessment, and personalized learning pathways (Naseer et al., 2024; Nhan, 2025). AI-enabled systems offer customized adaptive learning paths and real-time comparison between performance and sheet music, keeping practice both challenging and accurate (Yu, 2021; Huang & Ding, 2022; Li, 2022).

AI shows promise for making piano learning more efficient and personalized. Yet reviews also reveal troubling gaps: we still lack clarity about effectiveness boundaries, equity concerns, and how teaching methods—along with teachers themselves—must evolve (Merchán Sánchez-Jara et al., 2024). These are not peripheral questions; they go to the heart of what responsible AI integration in artistic education means. Through systematic review and meta-analytic methodology, this study offers the first quantitative synthesis of effect sizes in AI-enhanced piano education, establishing empirical benchmarks (pooled $d = 0.442$) for evaluating intervention effectiveness across technical skills, musical expression, and learning motivation.

Given this research background, this study conducts a systematic review and meta-analysis of research on AI-enhanced piano education from 2020 to 2025, focusing on three areas:

- The impact of AI on piano learning outcomes
- Ethical risks and governance recommendations arising from AI in piano education
- The transformation of teachers' roles and professional development driven by AI

Through systematic review and meta-analytic methodology, the author seeks to offer an evidence-based foundation for piano education practice, teacher training, and related policy development, while identifying future research directions. Research questions include:

- RQ1: How does AI influence piano students' technical proficiency, expressiveness, and learning motivation in existing studies?
- RQ2: What ethical and governance issues related to AI-enhanced piano education are addressed in current literature?
- RQ3: How does the integration of AI into piano instruction reshape the teacher's role? What are their requirements for digital literacy and AI literacy?

2. Methodology

2.1. Research Design

This study used a systematic review with meta-analytic synthesis to examine empirical research on AI use in piano education published between 2020 and 2025. The review followed PRISMA 2020 guidelines, which offer an internationally recognized standard for transparent reporting of systematic reviews (Page et al., 2021). The systematic review component sought to map dominant research themes, instructional approaches, ethical considerations, and reported changes in teachers' professional roles within the selected literature, consistent with prior reviews in educational technology and music education research (Bond et al., 2024; Zawacki-Richter et al., 2019). The qualitative synthesis was paired with a quantitative meta-analytic examination of effect size distributions related to AI-supported interventions in piano education. This component estimated pooled effect sizes and examined variability and patterns of reported effects on learning outcomes, including technical proficiency, musical expressiveness, and learning motivation. Such an integrative approach is widely used in educational research to synthesize evidence across heterogeneous study designs and instructional contexts, particularly in emerging technology domains (Slavin, 1986; Borenstein et al., 2009; Ahn et al., 2012).

For cross-study comparison, reported quantitative outcomes were standardized using Cohen's *d* as the primary effect size metric, with accompanying 95% confidence intervals (CIs). When effect sizes were not explicitly reported, estimates were derived from available statistical information (e.g., group means and standard deviations or test statistics) following established meta-analytic conversion procedures (Lipsey & Wilson, 2001; Borenstein et al., 2009). In studies using pre-post designs without control groups,

standardized mean gain scores were calculated where sufficient descriptive statistics were available. Given the diversity of participant characteristics, AI technologies, instructional designs, and outcome measures across the included studies, effect sizes were synthesized using a random-effects meta-analytic model, acknowledging that true effects may vary across educational contexts (Raudenbush, 2009; Higgins et al., 2019). This approach yields more conservative estimates than fixed-effect models and is recommended when substantial between-study heterogeneity is expected (Borenstein et al., 2009).

The integration of systematic qualitative synthesis with quantitative meta-analytic procedures offers a methodologically grounded framework for examining the educational efficacy, ethical implications, and pedagogical consequences of AI integration in piano education, while maintaining appropriate analytical caution in interpreting heterogeneous empirical evidence.

2.2. Search Strategy and Data Sources

Based on the research quality assessment framework proposed by Yang and Welch (2023), the journal articles selected for this review were evaluated according to specific criteria related to reliability, verifiability, credibility, and perceived transferability. To ensure comprehensive coverage of peer-reviewed literature, we searched the following databases: Web of Science, Scopus, and ProQuest.

A systematic search was conducted using Boolean operators and keywords designed to capture AI-enabled piano education literature. Examples include:

Web of Science (WoS):((TS=("Artificial Intelligence" OR "AI" OR "Machine Learning" OR "Deep Learning" OR "Intelligent System" OR "Adaptive Learning")) AND TS=("Piano Education" OR "Music Education" OR "Keyboard Learning" OR "Music Pedagogy"))AND TS=("Effectiveness" OR "Efficacy" OR "Outcome" OR "Performance" OR "Teacher Role" OR "Instructor Role" OR "Ethics" OR "Ethical Issue"OR "Ethical Issue" OR "Transformation" OR "Change"))

Scopus:(TITLE-ABS-KEY ("Artificial Intelligence" OR AI OR "Machine Learning" OR "Deep Learning" OR "Intelligent System" OR "Adaptive Learning")) AND (TITLE-ABS-KEY ("Piano Education" OR "Music Education" OR "Keyboard Learning" OR "Music Pedagogy")) AND (TITLE-ABS-KEY ("Effectiveness" OR Efficacy OR Outcome OR Performance OR "Teacher Role" OR "Instructor Role" OR Ethics OR "Ethical Issue" OR Transformation OR Change))

ProQuest:(TITLE-ABS-KEY ("Artificial Intelligence" OR AI OR "Machine Learning" OR "Deep Learning" OR "Intelligent System" OR "Adaptive Learning")) AND (TITLE-ABS-KEY ("Piano Education" OR "Music Education" OR "Keyboard Learning" OR "Music Pedagogy")) AND (TITLE-ABS-KEY ("Effectiveness" OR Efficacy OR Outcome OR Performance OR "Teacher Role" OR "Instructor Role" OR Ethics OR "Ethical Issue" OR Transformation OR Change))

Searches were applied to titles, abstracts, and keywords. Filters for publication year (2020–2025) and document type (article, review) were applied. Reference lists of key articles were also screened to ensure

completeness.

2.3. Study Selection and Screening

Table 1
Eligibility Criteria

Criteria	Description
Inclusion Criteria	I1: The study topic involves the application of AI or LLM in piano or broader music education. I2: The paper type is Empirical Study, Systematic Literature Review (SLR), high-quality Review, or critical conceptual article. I3: Published between January 1, 2020 and December 31, 2025. I4: The article provides substantive insights on core themes such as effectiveness, ethics, equity, or the transformation of the teacher’s role.
Exclusion Criteria	E1: Purely technical reports lacking educational or psychological discussion. E2: Conference abstracts, book reviews, or theses without full-text access (only a few peer-reviewed high-quality preprints are included). E3: Applications unrelated to music education (e.g., pure music information retrieval or music psychology experiments not involving teaching contexts).

By searching article titles, keywords, and abstracts, a total of 391 articles were identified. 88 duplicate records were removed; 60 records were marked as ineligible by automated tools; and 30 records were removed for other reasons, leaving 213 articles. After reviewing the titles and abstracts, 68 articles were excluded, leaving 145 articles. Five articles were deleted due to reasons such as temporary retraction, leaving 140 articles. Through full-text reading, 45 articles were excluded due to low quality; 28 articles were excluded due to contradictory data or small sample size; As shown in Fig. 1, a total of 62 articles were ultimately included in the final analysis.

This figure illustrates the classification of screened studies across two analytical dimensions: study quality and empirical validity/reliability. Only studies located in the upper-right quadrant, representing empirically robust and methodologically reliable research, were retained for meta-analysis.

2.4 Quality Assessment and Eligibility Classification

To rigorously evaluate the retrieved literature, we mapped the screening process onto a Cartesian coordinate system. See Fig. 2, which is defined by two dimensions: topic validity (vertical axis) and methodological reliability (horizontal axis). The initial 391 records from Web of Science, Scopus, and ProQuest covered a broad range. In the crucial full-text evaluation phase, we identified a unique cluster (top left quadrant) containing 78 studies; these studies, while exhibiting high topic relevance, were excluded due to methodological flaws, contradictory data, or insufficient rigor. Ultimately, only 62 empirical studies (right ellipse) that met both the high validity and high reliability criteria "survived" the metadata analysis and were included in the final systematic review.

2.5. Data Extraction and Coding

A standardized data extraction and coding protocol was developed to ensure methodological consistency and analytical reliability across the included studies (N = 62). For each eligible article, bibliographic information (author(s), year of publication, country/region, and journal source) was first recorded. Core methodological characteristics were then extracted, including sample size, participant characteristics (educational level and learning context), research design, and type of AI-based intervention employed in piano instruction (e.g., intelligent tutoring systems, performance analysis tools, adaptive feedback systems, or generative AI applications).

To facilitate systematic comparison, instructional outcomes were coded into three analytically distinct domains based on prevailing frameworks in music education and educational psychology:

- Technical skills, encompassing pitch accuracy, rhythmic stability, motor coordination, and sight-reading performance
- Musical expression, including dynamics control, expressive timing, articulation, and interpretive fluency
- Learning motivation, covering engagement, persistence, self-regulated practice, and affective responses toward piano learning

For quantitative synthesis, all reported outcomes were standardized to Cohen's *d* to enable cross-study comparability. When effect sizes were not explicitly reported, Cohen's *d* values were estimated using available statistical information (e.g., means and standard deviations, *t* values, *F* statistics, or *p* values) following established meta-analytic conventions (Lipsey & Wilson, 2001; Borenstein et al., 2009). Each effect size was accompanied by a 95% confidence interval (CI) to reflect estimation precision. In cases where multiple effect sizes were derived from a single study, each estimate was retained but treated as an independent analytic unit at the descriptive level to preserve domain-specific information. This extraction and coding procedure provided the empirical foundation for subsequent meta-analytic synthesis and visual representation through forest plots.

2.6. Meta-analytic Procedures

Following data extraction and standardization, a random-effects meta-analysis was conducted to synthesize effect size evidence across the included studies. Given the substantial heterogeneity in study designs (e.g., randomized controlled trials, quasi-experimental designs, pre-post comparisons), participant populations (age ranges from preschool to university students), AI intervention types (intelligent tutoring systems, adaptive feedback platforms, multimodal assessment tools), and outcome measures (pitch accuracy, expressive timing, practice engagement), a random-effects model was deemed most appropriate (Borenstein et al., 2009; Raudenbush, 2009). This modeling approach assumes that true effect sizes vary across studies due to differences in contextual and methodological factors, and provides more conservative estimates than fixed-effect models.

All extracted effect sizes were standardized to Cohen's d with corresponding 95% confidence intervals (CIs) to enable cross-study comparability. When effect sizes were not explicitly reported in the original studies, Cohen's d values were calculated from available statistical information (e.g., means and standard deviations, t -values, F -statistics) following established conversion procedures (Lipsey & Wilson, 2001; Borenstein et al., 2009). For studies reporting multiple outcomes within the same domain, effect sizes were averaged prior to inclusion in the meta-analysis to maintain statistical independence of effect size estimates.

Meta-analytic synthesis proceeded in three stages. First, an overall pooled effect size was calculated across all 62 studies to estimate the general magnitude of AI's impact on piano learning outcomes. Second, subgroup analyses were conducted to examine whether effect sizes differed systematically across instructional domains: technical skills (e.g., pitch accuracy, rhythmic precision, motor coordination), musical expression (e.g., dynamic control, expressive timing, articulation), and learning motivation (e.g., engagement, persistence, practice self-regulation). Third, heterogeneity was quantified using the Q statistic and I^2 index, with I^2 values of 25%, 50%, and 75% interpreted as low, moderate, and high heterogeneity, respectively (Higgins & Thompson, 2002).

To assess potential publication bias, visual inspection of funnel plots was conducted, supplemented by Egger's regression test (Egger et al., 1997). Asymmetry in the funnel plot or a statistically significant Egger's test ($p < .05$) would indicate possible publication bias, such as the selective reporting of studies with positive findings.

All meta-analytic procedures were performed using the *metafor* package (version 3.8-1) in R (version 4.3.1), following current best practices in educational meta-analysis (Pigott, 2012; Harrer et al., 2021). Forest plots were generated to provide visual representation of individual study effect sizes, confidence intervals, and the pooled estimate. While pooled effect sizes offer a quantitative summary of the literature, interpretation remained attentive to the diversity of instructional contexts, AI technologies, and pedagogical designs represented across studies. Accordingly, effect size estimates were treated as empirical benchmarks that support evidence-informed decision-making, while recognizing the importance of contextual factors in determining intervention effectiveness (Ahn et al., 2012; Reeves & Lin, 2020; Bond et al., 2024).

3. Result

3.1 Temporal Distribution of Publications (2020–2025)

The temporal distribution of the selected literature ($N = 62$) reveals a non-linear pattern of publication output over the six years from 2020 to 2025. As shown in Fig. 3, the number of studies remained relatively stable between 2020 and 2022, followed by a marked increase in 2023, during which publication output peaked at 16 studies. After this peak, the number of publications declined in 2024 and further decreased to eight studies in 2025.

To quantify the overall development of the field across the observation period, the compound annual growth rate (CAGR) was calculated based on publication counts from 2020 to 2025. Despite the substantial fluctuation observed in 2023, the overall CAGR was estimated at 2.71%, indicating a relatively modest net increase in publication output across the six years. This finding suggests that short-term surges in research activity were not sustained over time.

Overall, the observed temporal pattern can be characterized as a "rise-then-fall" trajectory, with a brief period of intensified publication activity followed by a return to lower output levels comparable to those observed at the beginning of the review period. These descriptive results provide a quantitative overview of the field's publication dynamics and serve as a contextual basis for subsequent analyses of research focus and empirical outcomes.

3.2. Meta-analytic Effect Size Synthesis

3.2.1 Overall Pooled Effect Size

Random-effects meta-analysis was conducted across all 62 included studies (N = 2,812 participants) to estimate the overall magnitude of AI-supported interventions on piano learning outcomes. As presented in Table 2, the pooled effect size was $d = 0.442$ (95% CI [0.398, 0.486], $p < .0001$), indicating a moderate positive effect according to conventional benchmarks (Cohen, 1988). This finding suggests that, on average, students receiving AI-enhanced piano instruction demonstrated learning gains approximately 0.44 standard deviations higher than comparison conditions (e.g., traditional instruction, control groups, or pre-intervention baselines).

Substantial between-study heterogeneity was observed ($Q = 156.78$, $df = 61$, $p < .0001$; $I^2 = 61.1\%$), indicating that approximately 61% of the observed variance in effect sizes reflected true differences across studies rather than sampling error alone (Higgins & Thompson, 2002). This level of heterogeneity is consistent with prior meta-analyses in educational technology research (Bond et al., 2024; Zawacki-Richter et al., 2019) and justified the use of random-effects modeling, which accounts for both within-study and between-study variance.

Table 2
Meta-analytic Summary of AI-Supported Piano Education Interventions

Domain	k	N	Cohen's d	95% CI	I ²	Q	p
Technical Skills	18	849	0.462	[0.389, 0.535]	57.8%	40.23	.002
Musical Expression	11	518	0.432	[0.339, 0.525]	61.7%	26.12	.006
Learning Motivation	33	1,445	0.434	[0.378, 0.490]	59.4%	78.81	< .001
Overall	62	2,812	0.442	[0.398, 0.486]	61.1%	156.78	< .0001

Note. k = number of studies; N = total participants; CI = confidence interval; I^2 = heterogeneity index (percentage of variance due to between-study heterogeneity); Q = Cochran's Q statistic for heterogeneity;

p = significance level. Effect sizes calculated using random-effects meta-analysis and interpreted using Cohen's (1988) benchmarks: small ($d = 0.20$), medium ($d = 0.50$), large ($d = 0.80$). Test of moderators indicated no significant differences between domains ($Q_{\text{between}} = 0.42$, $df = 2$, $p = .811$).

3.2.2 Subgroup Analysis by Instructional Domain

To examine whether AI's effectiveness varied systematically across different learning outcomes, subgroup analyses were conducted separately for technical skills, musical expression, and learning motivation (see Table 2 and Fig. 4).

Technical Skills ($k = 18$, $d = 0.462$, 95% CI [0.389, 0.535]): Studies examining technical outcomes such as pitch accuracy, rhythmic stability, fingering precision, and sight-reading performance yielded a moderate positive effect. The confidence interval did not overlap with zero, indicating reliable benefits for psychomotor skill development. Representative studies in this domain employed technologies such as automated performance evaluation systems (Phanichraksaphong & Tsai, 2021), real-time error detection (Yu, 2021), and adaptive difficulty adjustment (Tang, 2025).

Musical Expression ($k = 11$, $d = 0.432$, 95% CI [0.339, 0.525]): Effects for expressive dimensions—including dynamic control, tempo variation, articulation quality, and interpretive fluency—were comparable in magnitude to technical skills, contradicting the common assumption that AI is effective only for easily quantifiable outcomes. Studies in this category often utilized multimodal analysis (audio-visual fusion; Zhao et al., 2023), performance comparison with expert models (Giraldo et al., 2022), and visualized feedback on expressive parameters (Li, 2022). However, the relatively smaller number of studies ($k = 11$) and moderate heterogeneity ($I^2 = 61.7\%$) suggest this domain requires further empirical investigation.

Learning Motivation ($k = 33$, $d = 0.434$, 95% CI [0.378, 0.490]): AI-supported interventions showed moderate positive effects on motivational outcomes, including practice engagement, persistence, self-efficacy, and affective responses toward piano learning. This domain included the largest number of studies ($k = 33$), reflecting growing research interest in non-cognitive outcomes. Common intervention features included gamification elements (Huang & Ding, 2022), progress visualization (Deng et al., 2024), and adaptive challenge adjustment (Strielkowski et al., 2025). The consistency of positive effects across diverse motivational measures suggests that AI's capacity to provide immediate, personalized feedback may enhance learners' sense of competence and autonomy.

A test of moderators revealed no statistically significant differences between domains ($Q_{\text{between}} = 0.42$, $df = 2$, $p = .811$), indicating that effect sizes were comparable across technical, expressive, and motivational outcomes. The overlapping confidence intervals further support this interpretation. This finding challenges prevailing assumptions that AI's benefits are confined to lower-order technical skills, suggesting instead that well-designed AI systems can support multidimensional learning outcomes in piano education.

3.2.3. Effect Size Distribution and Visual Synthesis

Figure 4 presents a forest plot displaying individual study effect sizes in chronological order (2020–2025), along with 95% confidence intervals and the overall pooled estimate (indicated by the diamond marker). Visual inspection reveals that the majority of effect sizes (53 out of 62, or 85.5%) were positive, with only 9 studies reporting near-zero or slightly negative effects. The width of confidence intervals varied considerably across studies, reflecting differences in sample sizes (ranging from $N = 30$ to $N = 60$) and outcome measurement precision.

Several patterns emerge from the visual distribution. First, effect sizes clustered predominantly in the small-to-moderate range ($d = 0.20$ – 0.60), with few outliers exceeding $d = 0.70$. Second, studies published in 2023–2025 did not exhibit systematically larger or smaller effects compared to earlier years, suggesting temporal stability in reported outcomes. Third, confidence intervals for most studies overlapped substantially, indicating that while effect size point estimates varied, the precision of estimates was relatively consistent across the literature.

Note

This forest plot displays standardized effect sizes (Cohen's d) and 95% confidence intervals for all 62 included studies, organized chronologically by publication year. Each horizontal line represents a single study's effect size estimate and confidence interval. The vertical dashed line at $d = 0$ indicates no effect. Studies with confidence intervals not crossing the zero line demonstrate statistically significant effects. The diamond marker at the bottom represents the overall pooled effect size ($d = 0.442$, 95% CI [0.398 , 0.486]) calculated using random-effects meta-analysis. Color coding indicates instructional domain: blue = Technical Skills ($k = 18$), green = Musical Expression ($k = 11$), orange = Learning Motivation ($k = 33$). The width of confidence intervals reflects sample size and measurement precision, with narrower intervals indicating more precise estimates. Visual inspection reveals predominantly positive effects across all domains and years, with substantial between-study variability ($I^2 = 61.1\%$) supporting the use of random-effects modeling.

3.2.4. Publication Bias Assessment

Potential publication bias was assessed through visual inspection of a funnel plot (Fig. 5) and Egger's regression test. The funnel plot displayed a reasonably symmetric distribution of effect sizes around the pooled estimate, with studies dispersed across varying levels of standard error. Egger's regression test yielded a non-significant result ($z = 1.31$, $p = .191$), suggesting no strong evidence of asymmetry attributable to publication bias. This finding indicates that the observed pooled effect size is unlikely to be substantially inflated by the preferential publication of studies with positive results.

However, it is important to note that the absence of detectable publication bias does not guarantee its complete absence, particularly given that only peer-reviewed journal articles were included in this review. Studies with null or negative findings may remain unpublished or appear in non-indexed outlets (Pigott, 2012). Additionally, small-study effects—where smaller studies report larger effect sizes—were not strongly evident in this dataset, further supporting the robustness of the pooled estimate.

Publication Bias Assessment:

- Egger's Regression Test: $z = 1.31$, $p = .191$ (not significant)
- Pooled Effect Size: $d = 0.442$ (95% CI [0.398, 0.486])
- Number of Studies: $k = 62$
- Interpretation: No significant evidence of publication bias detected

Note

This funnel plot displays the relationship between effect sizes (Cohen's d , x-axis) and their standard errors (y-axis, inverted) for all 62 included studies. In the absence of publication bias, studies should be symmetrically distributed around the pooled effect estimate (vertical dashed line at $d = 0.442$), forming an inverted funnel shape. Studies with smaller sample sizes (larger standard errors) appear toward the top of the plot with greater horizontal dispersion, while larger studies (smaller standard errors) cluster more tightly near the pooled estimate at the bottom. Visual inspection reveals a reasonably symmetric distribution, with no pronounced asymmetry suggesting selective publication of positive findings. Egger's regression test confirmed this observation ($z = 1.31$, $p = .191$), indicating no significant evidence of publication bias. The triangular reference region (shaded) represents the 95% confidence interval around the pooled effect, within which 95% of studies would be expected to fall in the absence of heterogeneity.

3.2.5. Sensitivity Analysis

- To assess the robustness of the pooled effect size, sensitivity analyses were conducted by sequentially removing studies with the largest and smallest effect sizes. Removal of the three largest effect sizes ($d > 0.65$) resulted in a pooled estimate of $d = 0.425$ (95% CI [0.382, 0.468]), while removal of the three smallest effect sizes ($d < 0.25$) yielded $d = 0.450$ (95% CI [0.405, 0.495]). Both estimates remained within the moderate range and closely aligned with the original pooled effect ($d = 0.442$), indicating that the overall finding was not disproportionately influenced by extreme values.
- Additionally, a leave-one-out analysis was performed, in which the meta-analysis was re-run 62 times, each time excluding a different study. Pooled effect sizes ranged from $d = 0.435$ to $d = 0.449$, with all estimates remaining statistically significant ($p < .001$). This narrow range further supports the stability and reliability of the meta-analytic synthesis.

3.2.6. Summary of Meta-analytic Findings

- The meta-analytic synthesis across 62 studies provides convergent evidence that AI-supported piano instruction yields moderate positive effects on learning outcomes, with comparable benefits observed across technical skills, musical expression, and learning motivation. The consistency of effects across domains challenges the notion that AI is effective only for easily quantifiable technical training, suggesting instead that AI systems—when appropriately designed—can support holistic musical development. The substantial heterogeneity observed ($I^2 = 61.1\%$) underscores the

importance of contextual factors, including instructional design quality, learner characteristics, and technological affordances, in determining the effectiveness of AI interventions.

3.3. AI Empowers Piano Students' Technical Skills, Expressiveness, Learning Motivation

In recent years, with the application of AI technologies such as deep learning, automatic music transcription, multimodal analysis, and intelligent feedback systems in music education, more studies have begun to explore their mechanisms of action on piano learning effectiveness and motivation. Based on these studies, we can examine AI's impact from three dimensions: (1) basic performance skills, (2) musical expressiveness, and (3) learning motivation and participation.

3.3.1 .Technical Skills

Traditional piano instruction relies heavily on teacher supervision for developing basic performance skills—a model constrained by limited class time and infrequent feedback during independent practice (Cao, 2024; Zhang, 2025). AI technologies address these structural limitations by providing immediate, quantifiable feedback on technical aspects such as pitch accuracy, rhythmic stability, and fingering precision. Research consistently demonstrates AI's effectiveness in technical training. Automated evaluation systems can classify articulation types (legato vs. staccato) with accuracy exceeding 89% (Phanichraksaphong & Tsai, 2021), while real-time error detection enables targeted correction during practice (Yu, 2021). Perhaps more significantly, AI systems construct adaptive learning paths by analyzing error patterns and adjusting difficulty dynamically, allowing learners to focus on specific technical obstacles rather than repeatedly playing entire pieces (Strielkowski et al., 2025; Zhai & Xu, 2022). The precision and objectivity of AI feedback represents a clear advantage over human real-time evaluation in most technical domains (Lu, 2025). Large language models further enhance this capacity by intelligently matching learning resources to students' skill levels (Deng et al., 2024). However, these advantages remain concentrated in quantifiable dimensions. Long-term impacts on advanced musical expression and artistic judgment require verification through larger-sample longitudinal studies, suggesting AI functions better as an efficient auxiliary tool than as a replacement for teachers' aesthetic guidance.

3.3.2. Musical Expressiveness

Musical expressiveness—encompassing structural understanding, emotional communication, dynamic control, and stylistic interpretation—has long been considered resistant to technological intervention due to its subjective, context-dependent nature (Bonnaire & González-Moreno, 2023; Giraldo et al., 2022). Recent AI research challenges this assumption by decomposing expressiveness into algorithmically analyzable parameters: dynamic range profiles, inter-note timing, rhythmic elasticity, and articulation patterns. Multimodal approaches show particular promise. Audio-visual fusion models that jointly analyze acoustic effects and hand movements outperform audio-only systems in detecting expressive timing variations and interpretive intentions (Zhao et al., 2023). MIDI-based performance analysis

enables detailed modeling of dynamic contours and tempo variations, providing learners with quantitative feedback that clarifies abstract concepts like "layering" and "musical breath" more effectively than verbal instruction alone (Ru, 2025; Zhang, 2025). Even young learners show heightened attention to dynamic contrast and musical coherence when provided with visualized expressive feedback (Li, 2022). Yet significant limitations persist. Current AI systems focus primarily on measurable expressive parameters while struggling with emotional content, stylistic context, and cultural significance (Giraldo et al., 2022). Empirical research concentrates on short-term interventions with small samples, lacking longitudinal evidence on developmental trajectories and transfer effects (Yang et al., 2024). Moreover, over-reliance on standardized feedback may inadvertently constrain personalized interpretation and creative expression (Zhai et al., 2024). These constraints underscore that while AI can support expressive skill development, it cannot replace the nuanced aesthetic judgment central to artistic piano teaching.

3.3.3. Learning Motivation And Participation

Piano learning demands sustained autonomous practice, making intrinsic motivation and self-regulation critical for long-term achievement (Panadero, 2020). Traditional instruction maintains motivation primarily through teacher supervision and external rewards—mechanisms unavailable during independent practice when learners face vague goals and declining self-efficacy (Zhang, 2025). AI systems address these challenges by reconstructing practice experiences through immediate feedback, progress visualization, and adaptive challenge adjustment. Empirical evidence demonstrates that AI-enhanced environments significantly boost motivation, particularly among beginners. Gamification elements, instant scoring, and achievement tracking reduce practice frustration while enhancing engagement sustainability (Huang & Ding, 2022). In one study of preschool children, 84% reported that intelligent systems helped them practice proactively (Li, 2022). University-level blended learning incorporating AI feedback shows similar patterns: learners achieve technical improvements while demonstrating heightened engagement and persistence (Tang, 2025). Beyond motivation, AI supports self-regulated learning (SRL) by facilitating goal-setting, strategic practice, and self-monitoring. Rather than requiring students to repeatedly play entire pieces, AI identifies specific technical obstacles and delivers targeted practice segments, fostering more strategic learning behaviors (Zhai & Xu, 2022). Adaptive systems maintain learning within the "optimal challenge zone" by dynamically adjusting content based on real-time performance, promoting sustained engagement and self-efficacy development (Strielkowski et al., 2025). Emerging LLM-based platforms extend this support into reflective learning by generating personalized explanations and practice suggestions. Such dialogue-based feedback helps externalize previously implicit self-regulation processes, potentially enhancing metacognitive development (Deng et al., 2024; Jin et al., 2025). However, LLM feedback quality remains dependent on training data and prompt design, with artistic judgment and stylistic interpretation still requiring expert teacher guidance (Zhou et al., 2024).

3.4. Ethical Issues in Piano Education Empowered by AI

The proliferation of AI in piano instruction raises multiple ethical concerns requiring systematic attention. Foremost among these is data privacy: AI systems collect extensive audio, video, and behavioral data—including images of minors' home environments—creating substantial risks for misuse, re-identification, and unauthorized sharing absent robust governance frameworks (UNESCO, 2023; Right to Education Initiative, 2022). Copyright disputes present another governance challenge. AI training often relies on large-scale scores and recordings, many protected by copyright, as recent lawsuits demonstrate (The Guardian, 2023; Reuters, 2025). The ownership and usage boundaries of AI-generated accompaniments, adapted etudes, or teaching materials remain legally ambiguous. Algorithmic fairness concerns compound these issues: imbalanced training data may produce systematic biases against non-mainstream performance styles or non-Western musical traditions, potentially affecting learners' self-perception and educational opportunities (Baker, 2022). The "black box" nature of deep learning models undermines transparency and trust. Automated assessment systems that cannot explain their judgment criteria erode confidence among teachers, students, and parents—particularly problematic in arts education where evaluations directly shape learning trajectories (Miao & Holmes, 2021). Responsibility attribution remains unresolved: when AI produces erroneous outputs affecting students, determining whether developers, platforms, schools, or teachers bear accountability lacks consensus (Bell, 2024). Additional concerns include psychological impacts of continuous monitoring, the digital divide exacerbated by high-cost platforms, and risks of corporate control over teaching tools weakening institutional autonomy (UNESCO, 2023). Existing literature converges on a clear position: effective AI deployment in piano education demands explicit ethical guidelines, transparent data governance, and teacher-centered human-machine collaboration rather than technological substitution (Alam, 2021; Fitria, 2023; Zhao et al., 2021).

3.5. Transformation of the Teacher's Role

Empirical evidence reveals both strong acceptance and significant implementation barriers regarding AI in music teaching. Teachers express considerable willingness to adopt AI tools (mean = 4.24/5.00 on Likert scales; Aguila et al., 2024), recognizing how generative AI optimizes self-directed learning through immediate feedback (Li & Wang, 2024). However, substantial obstacles persist: cognitive burden during implementation, insufficient professional training, and concerns about algorithmic bias limiting pedagogical autonomy (Atabek & Burak, 2024; Kehoe, 2023).

Current generative models exhibit cultural biases that threaten both originality and equity in music education (Amankwah-Amoagh et al., 2024). While AI-generated lesson plans may demonstrate high quality, they lack context-adaptability and require extensive human intervention to achieve pedagogical effectiveness (Kehoe, 2023). These findings point toward a reconfiguration rather than replacement of teachers' roles. Educators increasingly function as aesthetic mentors, pedagogical decision-makers, and technology coordinators rather than sole providers of technical instruction (Aguila et al., 2024; Bell, 2024). This evolution demands new competencies. Teachers require not only digital literacy but also critical understanding of AI capabilities and limitations, enabling them to mitigate algorithmic bias and navigate complex, context-dependent instructional processes (Atabek & Burak, 2024). The sustainability

of AI integration thus depends less on technological advancement than on comprehensive teacher professional development, clear ethical frameworks, and institutional support structures that preserve pedagogical agency while leveraging technological affordances.

4. Discussion

Drawing on empirical evidence from 62 studies of AI-related piano education published between 2020 and 2025, the present systematic review and meta-analysis reveals several critical findings regarding the educational efficacy, methodological rigor, and contextual validity of current research. The meta-analytic synthesis demonstrated that AI-supported interventions yield a moderate positive effect on piano learning outcomes ($d = 0.442$, 95% CI [0.398, 0.486]), with comparable benefits observed across technical skills, musical expression, and learning motivation. This quantitative finding provides the first pooled effect size estimate in the domain of AI-enhanced piano education and offers an empirical benchmark for future intervention studies and policy development. However, while the overall effect size indicates meaningful learning gains, substantial heterogeneity was observed ($I^2 = 61.1\%$), suggesting that the magnitude of AI's impact varies considerably across studies. This variability points to the critical importance of contextual factors—including instructional design quality, technological affordances, learner characteristics, and pedagogical integration strategies—in determining intervention effectiveness. Closer examination of the reviewed literature raises four major concerns related to research scope, contextual validity, methodological balance, and pedagogical alignment, which collectively affect the interpretation, generalizability, and practical implications of current research findings (Chen, 2020; Zhai & Xu, 2022; Selwyn, 2023; Bell, 2024).

4.1. The Significance and Limitations of Moderate Effect Sizes

The observed pooled effect size of $d = 0.442$ falls within Cohen's (1988) classification of a "medium" effect and is comparable to meta-analytic findings in related educational technology domains. For instance, prior reviews of intelligent tutoring systems in mathematics education reported effect sizes ranging from $d = 0.35$ to $d = 0.50$ (Steenbergen-Hu & Cooper, 2014), while meta-analyses of computer-assisted instruction across disciplines yielded similar estimates (Cheung & Slavin, 2013). This convergence suggests that AI's impact on piano learning aligns with broader patterns observed in technology-enhanced education, where moderate positive effects are typical for well-designed interventions.

Importantly, the finding that effect sizes were comparable across technical skills ($d = 0.462$), musical expression ($d = 0.432$), and learning motivation ($d = 0.434$) challenges a prevalent assumption in music education discourse: that AI's benefits are confined to easily quantifiable, lower-order technical outcomes. The near-equivalent effects for musical expression—an inherently interpretive and context-dependent dimension of musicianship—suggest that AI systems, when designed with pedagogical sophistication, can support higher-order learning processes. This may reflect recent advances in

multimodal AI technologies (e.g., audio-visual fusion models; Zhao et al., 2023) and performance analysis tools that provide learners with visualized feedback on expressive parameters such as dynamic contours and tempo variation (Li, 2022; Giraldo et al., 2022).

Nevertheless, several critical limitations must be acknowledged. First, the moderate effect size reflects short-term outcomes in the majority of studies (median intervention duration: 8–12 weeks), leaving long-term retention, skill transfer, and sustained motivational effects largely unexamined. Second, the high heterogeneity ($I^2 = 61.1\%$) indicates that not all AI interventions are equally effective, and some studies reported near-zero or even slightly negative effects (9 out of 62 studies). This variability underscores that technological sophistication alone does not guarantee pedagogical value; rather, effectiveness depends on alignment between AI affordances and instructional goals, as well as on thoughtful integration into existing teaching practices (Holmes et al., 2022).

Third, while the absence of significant publication bias (Egger's test: $p = .191$) is reassuring, the exclusive focus on peer-reviewed journal articles may still introduce selection bias, as studies with null or negative findings are less likely to be published (Pigott, 2012). The meta-analytic estimate should therefore be interpreted as an upper-bound approximation of AI's impact under favorable implementation conditions, rather than as a definitive causal claim applicable to all contexts.

4.2. Technology-Driven Orientation and Pedagogical Implications

A strong technology-driven orientation was identified across the majority of reviewed studies. Much of the existing research frames AI as a solution to long-standing challenges in piano education, particularly those associated with limited teacher feedback, inefficient practice, and subjective assessment (Chen, 2020; Zhai & Xu, 2022). This optimism aligns with broader educational technology discourses that emphasize innovation, automation, and efficiency (Holmes et al., 2022). However, as observed in policy-driven educational reforms in music education more broadly, such technological enthusiasm may obscure deeper pedagogical questions. In many studies, the integration of AI systems is treated as a neutral or inherently beneficial intervention, while underlying assumptions about musicianship, artistic judgment, and learning processes remain insufficiently theorized (Bell, 2024; Selwyn, 2023). For example, several studies equate "improved learning outcomes" with higher scores on automated pitch detection or rhythm accuracy tests (Yu, 2021; Phanichraksaphong & Tsai, 2021), without addressing whether these gains translate into more musically meaningful performances or whether students develop interpretive autonomy. The meta-analytic finding of comparable effect sizes across domains (technical, expressive, motivational) suggests potential, but the lack of qualitative investigation into how students experience and make sense of AI-mediated feedback limits our understanding of these processes.

As a result, there appears to be a growing gap between technologically sophisticated research outputs and the nuanced realities of piano teaching practices, particularly with regard to expressive interpretation, aesthetic judgment, and the relational dimensions of teacher–student interaction. This gap is reflected in the reviewed studies' predominant reliance on short-term, quantifiable outcome

measures (e.g., pitch accuracy rates, completion times) while marginalizing dimensions of musicianship that resist standardization—such as stylistic sensitivity, emotional communication, and creative risk-taking (Bell, 2024).

4.3. The Illusion of Scalability: Contextual Validity and Educational Equity

The literature exhibits a tendency to frame increased technological investment as a primary means of improving piano education quality. Advanced AI systems—such as intelligent pianos, multimodal performance analysis tools, and adaptive learning platforms—are frequently presented as scalable solutions capable of transforming learning outcomes across diverse contexts (Li, 2022; Strielkowski et al., 2025). While the meta-analytic findings support the claim of positive average effects in controlled settings, they risk oversimplifying the complex conditions under which piano education occurs. A critical examination of the reviewed studies reveals that a substantial proportion of empirical research was conducted in economically developed regions, elite institutions, or technology-rich environments where access to AI-supported tools is readily available (Zhai & Xu, 2022; Strielkowski et al., 2025). For instance, 68% of the included studies were conducted in East Asia (particularly China, South Korea, Taiwan) and Western Europe, with only 12% originating from Latin America, Africa, or Southeast Asia. Moreover, 73% of studies involved participants from university-affiliated music programs, conservatories, or private music schools—settings characterized by high levels of institutional resources and technological infrastructure. This concentration suggests a form of contextual and selection bias, whereby reported benefits may reflect favorable implementation conditions rather than universally applicable outcomes (Baker et al., 2022). The external validity of findings is further constrained by limited attention to factors such as:

Digital access disparities: Few studies addressed whether students had reliable internet connectivity, personal devices, or adequate home practice spaces for AI-supported learning. Socioeconomic diversity: Sample descriptions rarely included information on participants' socioeconomic backgrounds, making it difficult to assess whether AI interventions exacerbate or mitigate existing educational inequalities. Cultural and linguistic contexts: AI systems trained predominantly on Western classical music repertoire and pedagogical norms may exhibit algorithmic bias when applied to non-Western musical traditions or pedagogies (Baker, 2022; Amankwah-Amoagh et al., 2024).

As with broader concerns in educational research, such bias limits the transferability of findings to under-resourced contexts or regions with limited digital infrastructure (UNESCO, 2023). The moderate pooled effect size ($d = 0.442$) should therefore be interpreted with caution, recognizing that it may overestimate AI's impact in less advantaged educational settings.

4.4. Methodological Imbalance: What Gets Measured, What Gets Marginalized

A fourth concern relates to a methodological imbalance between technical performance metrics and broader educational outcomes. The meta-analysis revealed that many studies prioritized quantifiable indicators—such as pitch accuracy, rhythm precision, or error rates—while comparatively few examined long-term learning trajectories, self-regulated practice behaviors, teacher professional development, or ethical implications (Panadero, 2020; Zhai & Xu, 2022). This imbalance is reflected in the distribution of outcome measures across the 62 studies:

- Technical accuracy metrics: 73% of studies (k = 45)
- Practice engagement/time-on-task: 53% of studies (k = 33)
- Musical expressiveness (subjective assessment): 18% of studies (k = 11)
- Long-term retention (> 3 months post-intervention): 6% of studies (k = 4)
- Teacher perspectives or professional development: 8% of studies (k = 5)

The dominance of short-term experimental designs and system validation studies may inadvertently reinforce a narrow conception of learning success, marginalizing aspects of musicianship that resist automation or standardization—such as interpretive originality, stylistic flexibility, and the capacity for critical aesthetic judgment (Bell, 2024). While the meta-analysis found comparable effect sizes for musical expression ($d = 0.432$), this finding is based on only 11 studies, many of which operationalized "expressiveness" through relatively objective parameters (e.g., dynamic range profiles, tempo deviation statistics) rather than holistic artistic judgments. This methodological trend parallels broader critiques in critical music education scholarship, which caution against reducing musicianship to measurable outcomes (Regelski, 2002). If AI research continues to prioritize easily quantifiable indicators, there is a risk of implicitly valuing technical compliance over creative exploration, and efficiency over aesthetic depth—ultimately shaping a narrower vision of what it means to learn piano (Selwyn, 2023).

4.5. Toward Context-Sensitive, Pedagogically Grounded AI Integration

Taken together, these findings suggest that while AI holds genuine promise for supporting piano education, its educational value cannot be realized through technological advancement alone. The moderate positive effect size ($d = 0.442$) observed in this meta-analysis should be understood not as definitive proof of AI's superiority, but as evidence that well-designed, pedagogically grounded AI systems can meaningfully augment certain dimensions of piano learning when implemented under favorable conditions. Future research should adopt more context-sensitive, theory-informed, and pedagogically nuanced approaches that recognize:

- The irreplaceable role of teachers in fostering artistic judgment, interpretive autonomy, and emotional connection to music
- The diversity of learning environments, including under-resourced schools, community music programs, and non-Western pedagogical traditions

- The artistic dimensions of piano education that extend beyond measurable technical proficiency, encompassing creativity, stylistic sensitivity, and musical meaning-making

Without such recalibration, there remains a risk that AI-enhanced piano education research may offer a technologically refined but pedagogically partial account of educational improvement—one that privileges efficiency and scalability over the complexity and depth that characterize meaningful musical learning (Holmes et al., 2022; UNESCO, 2023).

5. Conclusions and Future Research Directions

5.1. Conclusions

Our systematic review and meta-analysis of 62 studies (N = 2,812 participants) spanning 2020 through 2025 establishes the first empirical benchmark for AI in piano education. The central finding—a moderate positive effect ($d = 0.442$, 95% CI [0.398, 0.486], $p < .0001$)—tells an encouraging yet nuanced story about AI's potential. What surprised us most was the consistency across domains. Effect sizes for technical skills ($d = 0.462$), musical expression ($d = 0.432$), and learning motivation ($d = 0.434$) proved remarkably similar, challenging conventional wisdom that AI helps only with measurable technical drills. The comparable impact on musical expressiveness—an inherently interpretive dimension—suggests that thoughtfully designed systems can indeed support multidimensional learning when properly integrated into pedagogical contexts. However, the substantial heterogeneity observed ($I^2 = 61.1\%$) complicates any straightforward endorsement. Not all AI interventions work equally well. Some studies reported near-zero effects while others found substantial gains, underscoring that technological sophistication alone guarantees nothing. Effectiveness hinges on instructional design quality, learner characteristics, and alignment between AI affordances and pedagogical goals (Holmes et al., 2022; Selwyn, 2023). The evidence reveals both promise and limitations. AI-supported tools effectively enhance short-term technical accuracy, deliver immediate feedback, and increase practice engagement (Chen, 2020; Li, 2022; Zhao et al., 2023). Yet our review also exposes methodological imbalances: most studies prioritize quantifiable performance indicators while marginalizing dimensions like stylistic sensitivity, emotional communication, and creative risk-taking. Although we found comparable effect sizes for musical expression, this rests on only 11 studies, many operationalizing expressiveness through objective parameters rather than holistic artistic judgment. Ethical concerns emerged repeatedly across the literature. AI piano systems rely heavily on learner data—audio, video, behavioral logs—raising privacy risks especially acute when minors are involved (UNESCO, 2023). Algorithmic bias threatens to disadvantage learners from diverse musical backgrounds, as training data predominantly reflects Western classical traditions (Baker et al., 2022; Amankwah-Amoagh et al., 2024). The concentration of empirical research in well-resourced institutions compounds questions about generalizability to community music programs or non-Western pedagogical traditions.

Importantly, our analysis indicates reconfiguration rather than replacement of teachers' roles. Educators increasingly serve as aesthetic mentors, pedagogical strategists, and technology coordinators rather

than mere technical instructors (Aguila et al., 2024; Bell, 2024). While teachers express high AI acceptance (mean = 4.24/5.00), they report substantial implementation barriers: cognitive burden, insufficient training, and concerns about algorithmic bias (Atabek & Burak, 2024). This suggests sustainability depends not merely on technological development but on comprehensive professional development, clear ethical frameworks, and institutional support preserving pedagogical agency.

In summary, AI holds genuine potential for supporting piano learning, but its educational value depends critically on pedagogically informed integration rather than technological capability alone. The moderate effect size ($d = 0.442$) should be understood not as proof of AI's superiority but as evidence that well-designed systems, when implemented thoughtfully and equitably, can meaningfully augment certain dimensions of instruction. Advancing the field requires context-sensitive, theory-driven, ethically informed approaches recognizing teachers' irreplaceable role, environmental diversity, and the artistic foundations distinguishing musical from purely technical education.

5.2. Research Gaps and Future Research Directions

Our review identifies five critical gaps warranting empirical attention:

First, longitudinal evidence remains scarce. Most studies examine short-term outcomes (median duration: 8–12 weeks), leaving sustained retention, skill transfer, and long-term motivational effects largely unexamined. Future research should adopt longitudinal designs tracking how AI-supported practice influences technical development, expressive growth, and self-regulated learning over extended periods (Panadero, 2020; Zhai & Xu, 2022).

Second, theoretical grounding needs strengthening. Many studies focus on measurable indicators while remaining weakly connected to established learning theories. Explicitly integrating frameworks like self-regulated learning, experiential learning, or constructivist pedagogy would clarify mechanisms through which AI supports learning (Holmes et al., 2022; Bell, 2024).

Third, contextual diversity and equity require systematic investigation. Evidence concentrates in economically developed regions and elite institutions. Comparative research across diverse socio-economic contexts is essential for evaluating feasibility, adaptability, and inclusiveness of AI-enhanced instruction (Baker et al., 2022; UNESCO, 2023).

Fourth, teachers' professional experiences need deeper exploration. While shifts toward facilitation and learning design are acknowledged, few studies investigate how educators negotiate professional identity, pedagogical authority, and ethical responsibility within AI-mediated environments. Teacher-centered qualitative research is critical for understanding sustainable integration models (Selwyn, 2023; Bell, 2024).

Finally, ethical governance in piano-specific applications remains underexplored. Issues like data privacy, algorithmic transparency, bias, and accountability appear in general AI-in-education literature but rarely

receive empirical examination in music education. Future research should develop operational frameworks ensuring responsible, trustworthy AI deployment (Baker et al., 2022; UNESCO, 2023).

5.3. Concluding Remarks

In conclusion, while artificial intelligence offers promising tools for enhancing aspects of piano education, its pedagogical impact cannot be reduced to technological efficiency or performance optimization. Advancing the field requires a shift toward context-sensitive, theory-driven, and ethically informed research agendas that recognize the irreplaceable role of teachers, the diversity of learning environments, and the artistic foundations of piano education. Addressing these challenges will be essential for ensuring that AI-enhanced piano education contributes meaningfully to both educational quality and musical integrity.

References

1. Aguila AA, Al-Samarraie H, Al-Buraiki A, Al-Sharhan N, Aldheleai T (2024) Artificial intelligence in teaching music skills and teachers' attitudes towards it in the Sultanate of Oman. *J Eng Appl Sci* 19(5):1–12. <https://doi.org/10.36478/jeasci.2024.19.05>
2. Ahn S, Ames AJ, Myers ND (2012) A review of meta-analyses in education: Methodological strengths and weaknesses. *Rev Educ Res* 82(4):436–476. <https://doi.org/10.3102/0034654312458162>
3. Akgun S, Greenhow C (2022) Artificial intelligence in education: Addressing ethical challenges in K-12 settings. *AI Ethics* 2(3):431–440. <https://doi.org/10.1007/s43681-021-00096-7>
4. Alam A (2021), December Should robots replace teachers? Mobilisation of AI and learning analytics in education. In *2021 International Conference on Advances in Computing, Communication, and Control (ICAC3)* (pp. 1–12). IEEE. <https://doi.org/10.1109/ICAC353642.2021.9697300>
5. Ali O, Murray PA, Momin M, Dwivedi YK, Malik T (2024) The effects of artificial intelligence applications in educational settings: Challenges and strategies. *Technol Forecast Soc Chang* 199., Article 123076. <https://doi.org/10.1016/j.techfore.2023.123076>
6. An X, Chai CS, Li Y, Zhou Y, Shen X, Zheng C, Chen M (2023) Modeling English teachers' behavioral intention to use artificial intelligence in middle schools. *Educ Inform Technol* 28(5):5187–5208. <https://doi.org/10.1007/s10639-022-11286-z>
7. Antonini Philippe R, Kosirnik C, Vuichoud N, Williamon A, von Roten FC (2020) Understanding wellbeing among college music students and amateur musicians in Western Switzerland. *Front Psychol* 11:820. <https://doi.org/10.3389/fpsyg.2020.00820>
8. Atabek O, Burak S (2024) Exploring pre-service music teachers' acceptance of generative artificial intelligence: A PLS-SEM-ANN approach. *Front Psychol* 15., Article 1571279. <https://doi.org/10.3389/fpsyg.2024.1571279>
9. Bahroun Z, Anane C, Ahmed V, Zacca A (2023) Transforming education: A comprehensive review of generative artificial intelligence in educational settings through bibliometric and content analysis.

- Sustainability 15(17):12983. <https://doi.org/10.3390/su151712983>
10. Bai B (2021) Piano learning in the context of schooling during China's 'piano craze' and beyond: Motivations and pathways. *Music Educ Res* 23(4):512–526. <https://doi.org/10.1080/14613808.2021.1929138>
 11. Baker RS, Hawn A, Zawacki-Richter O (2022) Algorithmic bias in education. *Int J Artif Intell Educ* 32(4):1055–1082. <https://doi.org/10.1007/s40593-021-00285-9>
 12. Banquero M, Valdeolivas G, Ramón D (2024) A color passthrough mixed reality application for learning piano. *Virtual Reality* 28(2). Article 67. <https://doi.org/10.1007/s10055-024-00967-8>
 13. Bell AP (2024) Artificial intelligence, automation, and the future of music education. *Action Criticism Theory Music Educ* 23(1):1–24. https://act.maydaygroup.org/articles/Bell23_1.pdf
 14. Bettayeb AM, Abu Talib M, Sobhe Altayasinah AZ, Dakalbab F (2024) Exploring the impact of ChatGPT: Conversational AI in education. *Front Educ* 9:1379796. <https://doi.org/10.3389/educ.2024.1379796>
 15. Bobbe T, Oppici L, Lüneburg LM, Meinel C (2021) What early user involvement could look like—Developing technology applications for piano teaching and learning. *Multimodal Technol Interact* 5(7). Article 38. <https://doi.org/10.3390/mti5070038>
 16. Bond M, Khosravi H, De Laat M, Buckingham Shum S, Gašević D, Sadiq S (2024) A meta-systematic review of artificial intelligence in education: Call for increased rigor, ethics, and theory. *Int J Educational Technol High Educ* 21., Article 4. <https://doi.org/10.1186/s41239-023-00436-z>
 17. Bonnaire S, González-Moreno P (2023) Developing metacognition as an instructional approach for enhancing young pianists' musical expressivity. *Int J Music Educ* 41(2):199–210. <https://doi.org/10.1177/02557614221097447>
 18. Borenstein J, Howard A (2021) Emerging challenges in AI and the need for AI ethics education. *AI Ethics* 1(1):61–65. <https://doi.org/10.1007/s43681-020-00002-7>
 19. Borenstein M, Hedges LV, Higgins JPT, Rothstein HR (2009) *Introduction to meta-analysis*. Wiley
 20. Cao Y (2024) Online courses for piano teaching: Comparing the effectiveness and impact of modern and traditional methods. *Educ Inform Technol* 29(9):11407–11419. <https://doi.org/10.1007/s10639-023-12225-0>
 21. Cardona MA, Rodríguez RJ, Ishmael K (2023) Artificial intelligence and the future of teaching and learning. U.S. Department of Education. <https://tech.ed.gov/files/2023/05/ai-future-of-teaching-and-learning-report.pdf> Office of Educational Technology
 22. Carnovalini F, Rodà A (2020) Computational creativity and music generation systems: An introduction to state of the art. *Front Artif Intell* 3., Article 14. <https://doi.org/10.3389/frai.2020.00014>
 23. Casal-Otero L, Catala A, Fernández-Morante C, Taboada M, Cebreiro B, Barro S (2023) AI literacy in K-12: A systematic literature review. *Int J STEM Educ* 10(1) Article 17. <https://doi.org/10.1186/s40594-023-00418-7>

24. Celik I (2023) Towards intelligent-TPACK: An empirical study on teachers' professional knowledge to ethically integrate artificial intelligence (AI)-based tools into education. *Comput Hum Behav* 138., Article 107468. <https://doi.org/10.1016/j.chb.2022.107468>
25. Chen JCW (2020) AI in music education: The impact of using artificial intelligence (AI) application to practise scales and arpeggios in a virtual learning environment. In *Learning environment and design: Current and future impacts* (pp. 307–322). Springer. https://doi.org/10.1007/978-981-15-8167-0_16
26. Chen X, Zou D, Cheng G, Xie H (2025) Exploring the impact of AI-assisted practice applications on music learners' performance, self-efficacy, and self-regulated learning. *Front Psychol* 15., Article 1675762. <https://doi.org/10.3389/fpsyg.2025.1675762>
27. Chen Y, Liu Y (2024) Unlocking the beat: How AI tools drive music students' motivation, engagement, creativity and learning success. *Eur J Educ* 59(4) Article e12823. <https://doi.org/10.1111/ejed.12823>
28. Chen Z (2023) Ethics and discrimination in artificial intelligence-enabled recruitment practices. *Humanit Social Sci Commun* 10., Article 567. <https://doi.org/10.1057/s41599-023-02079-x>
29. Cheng L (2025) The impact of generative AI on school music education: Challenges and recommendations. *Arts Educ Policy Rev* 126(2):1–8. <https://doi.org/10.1080/10632913.2025.2378654>
30. Cheung ACK, Slavin RE (2013) The effectiveness of educational technology applications for enhancing mathematics achievement in K-12 classrooms: A meta-analysis. *Educational Res Rev* 9:88–113. <https://doi.org/10.1016/j.edurev.2013.01.001>
31. Chiu TKF (2024) Future research recommendations for transforming higher education with generative AI. *Computers Education: Artif Intell* 6., Article 100197. <https://doi.org/10.1016/j.caeai.2024.100197>
32. Chiu TKF, Moorhouse BL, Chai CS, Ismailov M (2024) Teacher support and student motivation to learn with artificial intelligence (AI) based chatbot. *Interact Learn Environ* 32(6):3240–3256. <https://doi.org/10.1080/10494820.2023.2172044>
33. Choi S, Jang Y, Kim H (2023) Influence of pedagogical beliefs and perceived trust on teachers' acceptance of educational artificial intelligence tools. *Int J Hum Comput Interact* 39(5):910–922. <https://doi.org/10.1080/10447318.2022.2049145>
34. Civit M, Civit-Masot J, Cuadrado F, Escalona MJ (2022) A systematic review of artificial intelligence-based music generation: Scope, applications, and future trends. *Expert Syst Appl* 209:118190. <https://doi.org/10.1016/j.eswa.2022.118190>
35. Cohen J (1988) *Statistical power analysis for the behavioral sciences*, 2nd edn. Lawrence Erlbaum Associates
36. Cooper PK (2024) Music teachers' labeling accuracy and quality ratings of lesson plans by artificial intelligence (AI) and humans. *Int J Music Educ* 42(4):567–582. <https://doi.org/10.1177/02557614241249163>

37. Crompton H, Burke D (2023) Artificial intelligence in higher education: The state of the field. *Int J Educational Technol High Educ* 20(1) Article 22. <https://doi.org/10.1186/s41239-023-00392-8>
38. Cui K (2023) Artificial intelligence and creativity: Piano teaching with augmented reality applications. *Interact Learn Environ* 31(10):7017–7028. <https://doi.org/10.1080/10494820.2022.2073061>
39. Deja JA (2022) Piano learning and improvisation through adaptive visualisation and digital augmentation. In *Companion proceedings of the 2022 conference on interactive surfaces and spaces* (pp. 41–45). ACM. <https://doi.org/10.1145/3567729.3567736>
40. Della Ventura M (2023) Intelligent tutoring system and learning: Complexity and resilience. In M. Dascalu, Ó. Mealha, & S. Virkus (Eds.), *Smart learning ecosystems as engines of the green and digital transition* (pp. 25–38). Springer. https://doi.org/10.1007/978-981-99-5540-4_3
41. Deng W, Wang L, Deng X (2024) Strategies for optimizing personalized learning pathways with artificial intelligence assistance. *Int J Adv Comput Sci Appl* 15(6):234–245. <https://doi.org/10.14569/IJACSA.2024.0150626>
42. Deng Y, Lin Y (2025) The effect of ChatGPT on students' learning performance, learning perception, and higher-order thinking: Insights from a meta-analysis. *Humanit Social Sci Commun* 12., Article 4787. <https://doi.org/10.1038/s41599-025-04787-y>
43. Egger M, Davey Smith G, Schneider M, Minder C (1997) Bias in meta-analysis detected by a simple, graphical test. *BMJ* 315(7109):629–634. <https://doi.org/10.1136/bmj.315.7109.629>
44. European Commission (2022) DigCompEdu: The European framework for the digital competence of educators. Publications Office of the European Union. https://joint-research-centre.ec.europa.eu/digcompedu_en
45. Fitria TN (2023) The use of artificial intelligence in education (AIED): Can AI replace the teacher's role? *Epigram* 20(2):165–187. <https://doi.org/10.32722/epigram.v20i2.5443>
46. Förster A, Lepa S (2023) Digital musical instruments in special educational needs schools: Requirements from the music teachers' perspective and the status quo in Germany. *ACM Trans Accessible Comput* 16(3):29. <https://doi.org/10.1145/3616015>
47. Gándara D, Anahideh H, Ison M, Picchiarini L (2024) Inside the black box: Detecting and mitigating algorithmic bias across racialized groups in college student-success prediction. *AERA Open* 10(1):1–18. <https://doi.org/10.1177/23328584241258741>
48. Gao WX, Zhang S, Zhang NX (2023) Generating fingerings for piano music with model-based reinforcement learning. *Appl Sci* 13(20). Article 11321. <https://doi.org/10.3390/app132011321>
49. Gaunt H, Duffy C, Coric A, González Delgado IR, Messas L, Pryimenko O, Sveidahl H (2021) Musicians as 'makers in society': A conceptual foundation for contemporary professional higher music education. *Front Psychol* 12:713648. <https://doi.org/10.3389/fpsyg.2021.713648>
50. Gedik G, Demirtas S (2023) The relationship between self-directed learning levels and piano learning styles of music students. *Pamukkale Univ J Educ* 59:145–168. <https://doi.org/10.9779/pauefd.1196847>

51. Giraldo S, Cancino-Chacón C, Widmer G (2022) A review of computational models of expressive music performance. *ACM-CSUR* 54(3):1–37. <https://doi.org/10.1145/3464424>
52. Grájeda A, Córdova P, Córdova JP, Laguna-Tapia A, Burgos J, Rodríguez L, Sánchez K, Sanjinés A, Mamani P (2024) Embracing artificial intelligence in the arts classroom: Understanding student perceptions and emotional reactions to AI tools. *Cogent Educ* 11(1). Article 2378271. <https://doi.org/10.1080/2331186X.2024.2378271>
53. Hao J (2024) Online piano learning game design method: Piano music style recognition based on CRNNH. *Entertainment Comput* 50., Article 100645. <https://doi.org/10.1016/j.entcom.2024.100645>
54. Harrer M, Cuijpers P, Furukawa TA, Ebert DD (2021) Doing meta-analysis with R: A hands-on guide. Chapman & Hall/CRC. <https://doi.org/10.1201/9781003107347>
55. Hasanein AM, Sobaih AEE (2023) Drivers and consequences of ChatGPT use in higher education: Key stakeholder perspectives. *Eur J Invest Health Psychol Educ* 13(11):2599–2614. <https://doi.org/10.3390/ejihpe13110181>
56. Hatfield JL, Halvari H, Lemyre PN (2017) Instrumental practice in the contemporary music academy: A three-phase cycle of self-regulated learning in music students. *Musicae Sci* 21(3):316–337. <https://doi.org/10.1177/1029864917690206>
57. Henry A, Wiratama V, Afilipoaie A, Ranaivoson H, Arrivé E (2024) Impacts of AI on music consumption and fairness. *Music Sci* 7:1–15. <https://doi.org/10.1177/27523543241269047>
58. Higgins JPT, Thompson SG (2002) Quantifying heterogeneity in a meta-analysis. *Stat Med* 21(11):1539–1558. <https://doi.org/10.1002/sim.1186>
59. Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA (eds) (2019) *Cochrane handbook for systematic reviews of interventions* (2nd ed.). Wiley. <https://doi.org/10.1002/9781119536604>
60. Ho Weatherly KIC, Wright B, Lee EY (2024) Digital game-based learning in music education: A systematic review between 2011 and 2023. *Res Stud Music Educ* 46(3):456–478. <https://doi.org/10.1177/1321103X241270819>
61. Holmes W, Bialik M, Fadel C (2022) Artificial intelligence in education: Promise and implications for teaching and learning, 2nd edn. Center for Curriculum Redesign
62. Holmes W, Porayska-Pomsta K, Holstein K, Sutherland E, Baker T, Shum SB, Santos OC, Rodrigo MT, Cukurova M, Bittencourt II, Koedinger KR (2022) Ethics of AI in education: Towards a community-wide framework. *Int J Artif Intell Educ* 32(3):504–526. <https://doi.org/10.1007/s40593-021-00239-1>
63. Huang N, Ding X (2022) Piano music teaching under the background of artificial intelligence. *Wirel Commun Mob Comput* 2022(1) Article 5816453. <https://doi.org/10.1155/2022/5816453>
64. Hwang GJ, Xie H, Wah BW, Gašević D (2020) Vision, challenges, roles and research issues of artificial intelligence in education. *Computers Education: Artif Intell* 1:100001. <https://doi.org/10.1016/j.caeai.2020.100001>
65. Jeon J, Lee S (2023) Large language models in education: A focus on the complementary relationship between human teachers and ChatGPT. *Educ Inform Technol* 28(12):15873–15892.

<https://doi.org/10.1007/s10639-023-11834-1>

66. Jiang H (2024) Machine learning in teaching piano touch technique and tonal expression. *J Electr Syst* 20(4 SI):404–415
67. Jin L, Lin B, Hong M, Zhang K, So HJ (2025) Exploring the impact of an LLM-powered teachable agent on learning gains and cognitive load in music education. *arXiv preprint arXiv:2504.00636*. <https://doi.org/10.48550/arXiv.2504.00636>
68. Kalkanoğlu B (2025) The effect of using technology in music education and training on academic achievement: A meta-analysis study. *Educ Sci* 50(221):213–236. <https://doi.org/10.15390/EB.2024.13320>
69. Kandemir Ö, Yokuş T (2023) The effect of learning strategies on piano performance self-efficacy levels and performance success. *Cukurova Univ Fac Educ J* 52(2):446–470. <https://doi.org/10.14812/cuefd.1265516>
70. Karolus J, Sylupp J, Schmidt A (2023) EyePiano: Leveraging gaze for reflective piano learning. In *Proceedings of the designing interactive systems conference (DIS 2023)* (pp. 1209–1223). ACM. <https://doi.org/10.1145/3563657.3596122>
71. Kasneci E, Sessler K, Küchemann S, Bannert M, Dementieva D, Fischer F, Gasser U, Groh G, Günemann S, Hüllermeier E, Kruber S, Samuel A (2023) ChatGPT for good? On opportunities and challenges of large language models for education. *Learning and Individual Differences*, 103, Article 102274. <https://doi.org/10.1016/j.lindif.2023.102274>
72. Kehoe A (2023) AI and the music teacher: Navigating a new pedagogical landscape. *J Music Technol Educ* 16(1):45–62. https://doi.org/10.1386/jmte_00067_1
73. Konovalova I, Breslavets H, Riabukha N, Polska I, Shchepakina V, Roshchenko O (2025) The evolution of world music pedagogy in the information society. *BRAIN: Broad Res Artif Intell Neurosci* 16(1 Suppl):99–116. <https://doi.org/10.18662/brain/16.1Sup/929>
74. Lam CK (2024) Technology-enhanced creativity in K-12 music education: A scoping review. *Int J Music Educ* 42(1):89–108. <https://doi.org/10.1177/02557614231194073>
75. Li J, Wang Y (2024) Enhancing self-directed learning in music education through generative AI: A quantitative analysis. *Educ Inform Technol* 29(8):10234–10256. <https://doi.org/10.1007/s10639-023-12178-4>
76. Li S, Timmers R (2021) Teaching and learning of piano timbre through teacher-student interactions in lessons. *Front Psychol* 12., Article 576056. <https://doi.org/10.3389/fpsyg.2021.576056>
77. Li W (2022) Analysis of piano performance characteristics by deep learning and artificial intelligence and its application in piano teaching. *Front Psychol* 12., Article 751406. <https://doi.org/10.3389/fpsyg.2021.751406>
78. Lin CC, Huang AY, Lu OH (2023) Artificial intelligence in intelligent tutoring systems toward sustainable education: A systematic review. *Smart Learn Environ* 10(1). Article 41. <https://doi.org/10.1186/s40561-023-00260-y>
79. Lipsey MW, Wilson DB (2001) Practical meta-analysis. SAGE

80. Liu X, Shao X (2024) Modern mobile learning technologies in online piano education: Online educational course design and impact on learning. *Interact Learn Environ* 32(4):1279–1290. <https://doi.org/10.1080/10494820.2022.2118787>
81. Liu Y, Tran VC, Kiss B, Oo TZ, Szabó N, Józsa K (2025) The use and effectiveness of digital tools in elementary music education: A systematic review. *Music Educ Res* 27(1):45–68. <https://doi.org/10.1177/20592043251363338>
82. Long D, Magerko B (2020) What is AI literacy? Competencies and design considerations. In *Proceedings of the 2020 CHI conference on human factors in computing systems* (pp. 1–16). ACM. <https://doi.org/10.1145/3313831.3376727>
83. López-Íñiguez G, McPherson GE (2020) Applying self-regulated learning and self-determination theory to optimize the performance of a concert soloist. *Front Psychol* 11., Article 385. <https://doi.org/10.3389/fpsyg.2020.00385>
84. Lu L (2025) AI-powered intelligent music education systems for real-time feedback and performance assessment. *Int J Inf Commun Technol* 26(18):33–47
85. Luckin R, Holmes W, Griffiths M, Forcier LB (2016) Intelligence unleashed: An argument for AI in education. Pearson Educ. <https://www.pearson.com/corporate/about-pearson/innovation/smarter-digital-tools/intelligence-unleashed.html>
86. Ma R, Ma R (2022) Piano education online: Challenges and solutions. *Educ Inform Technol* 28:1359–1372. <https://doi.org/10.1007/s10639-022-11242-x>
87. MacRitchie J, Garrido S (2019) Measuring music performance self-efficacy in orchestral musicians. *Psychol Music* 47(6):856–870. <https://doi.org/10.1177/0305735618789572>
88. Matsui R, Kumaki M, Takegawa Y (2022) Proposal of a piano learning support system considering gamification based on reconsolidation. In *Learning and collaboration technologies: Designing the learner and teacher experience* (pp. 113–125). Springer. https://doi.org/10.1007/978-3-031-05657-4_9
89. McPherson GE, McCormick J (2006) Self-efficacy and music performance. *Psychol Music* 34(3):322–336. <https://doi.org/10.1177/0305735606064841>
90. Merchán Sánchez-Jara JF, González Gutiérrez S, Cruz Rodríguez J, Syroyid S, B (2024) Artificial intelligence–assisted music education: A critical synthesis of challenges and opportunities. *Educ Sci* 14(11):1171. <https://doi.org/10.3390/educsci14111171>
91. Miao F, Holmes W (2021) AI and education: Guidance for policy-makers. United Nations Educational. <https://unesdoc.unesco.org/ark:/48223/pf0000376709> Cultural Organization
92. Miksza P (2015) The effect of self-regulation instruction on the performance achievement, musical self-efficacy, and practicing of advanced wind players. *Psychol Music* 43(2):219–243. <https://doi.org/10.1177/0305735613500832>
93. Mishra P, Warr M, Islam R (2023) TPACK in the age of ChatGPT and generative AI. *J Digit Learn Teacher Educ* 39(4):235–251. <https://doi.org/10.1080/21532974.2023.2247480>

94. Naseer F, Khan MN, Tahir M, Addas A, Aejaaz SH (2024) Integrating deep learning techniques for personalized learning pathways in higher education. *Heliyon* 10(11) Article e32145. <https://doi.org/10.1016/j.heliyon.2024.e32145>
95. Ng DTK, Leung JKL, Chu SKW, Qiao MS (2021) Conceptualizing AI literacy: An exploratory review. *Computers and Education: Artificial Intelligence*, 2, Article 100041. <https://doi.org/10.1016/j.caeai.2021.100041>
96. Ng DTK, Leung JKL, Su J, Ng RCW, Chu SKW (2023) Teachers' AI digital competencies and twenty-first century skills in the post-pandemic world. *Education Tech Research Dev* 71(1):33–59. <https://doi.org/10.1007/s11423-023-10203-6>
97. Nhan LK (2025) The role of AI in enhancing personalised learning, automated assessment, intelligent tutoring, and student engagement. *Turkish Online J Distance Educ* 26(4):55–76. <https://doi.org/10.17718/tojde.1581237>
98. Niu Y (2021) Penetration of multimedia technology in piano teaching and performance based on complex network. *Math Probl Eng* 2021(1) Article 8872227. <https://doi.org/10.1155/2021/8872227>
99. Obermeyer Z, Powers B, Vogeli C, Mullainathan S (2019) Dissecting racial bias in an algorithm used to manage the health of populations. *Science* 366(6464):447–453. <https://doi.org/10.1126/science.aax2342>
100. Ouyang F, Jiao P (2021) Artificial intelligence in education: The three paradigms. *Computers and Education: Artificial Intelligence*, 2, Article 100020. <https://doi.org/10.1016/j.caeai.2021.100020>
101. Ouyang M (2023) Employing mobile learning in music education. *Educ Inform Technol* 28:5241–5257. <https://doi.org/10.1007/s10639-022-11353-5>
102. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, Shamseer L, Tetzlaff JM, Akl EA, Brennan SE, Chou R, Glanville J, Grimshaw JM, Hróbjartsson A, Lalu MM, Li T, Loder EW, Mayo-Wilson E, McDonald S, Moher D (2021) The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. *BMJ* 372 Article n71. <https://doi.org/10.1136/bmj.n71>
103. Panadero E (2020) A review of self-regulated learning: Six models and four directions for research. *Front Psychol* 11., Article 108. <https://doi.org/10.3389/fpsyg.2020.00108>
104. Phanichraksaphong V, Tsai W-H (2021) Automatic evaluation of piano performances for STEAM education. *Appl Sci* 11(7) Article 3042. <https://doi.org/10.3390/app11073042>
105. Pigott TD (2012) *Advances in meta-analysis*. Springer. <https://doi.org/10.1007/978-1-4614-2278-5>
106. Pintrich PR (1999) The role of motivation in promoting and sustaining self-regulated learning. *Int J Educational Res* 31(6):459–470. [https://doi.org/10.1016/S0883-0355\(99\)00015-4](https://doi.org/10.1016/S0883-0355(99)00015-4)
107. Raudenbush SW (2009) Analyzing effect sizes: Random-effects models. In: Cooper H, Hedges LV, Valentine JC (eds) *The handbook of research synthesis and meta-analysis*, 2nd edn. Russell Sage Foundation, pp 295–315
108. Reeves TC, Lin L (2020) The research we have is not the research we need. *Education Tech Research Dev* 68:1991–2001. <https://doi.org/10.1007/s11423-020-09811-3>

109. Regier BJ (2022) Development and validation of the jazz self-efficacy scale. *J Res Music Educ* 70(1):67–88. <https://doi.org/10.1177/00224294211054321>
110. Regelski TA (2002) On methodolatry and music teaching as critical and reflective praxis. *Philos Music Educ Rev* 10(2):102–123 [suspicious link removed]
111. Ritchie L, Williamon A (2011) Measuring distinct types of musical self-efficacy. *Psychol Music* 39(3):328–344. <https://doi.org/10.1177/0305735610374895>
112. Ru T (2025) Deep learning-based automatic piano note recognition and performance generation system. *Int J High Speed Electron Syst* 34(1) Article 2540814. <https://doi.org/10.1142/S0129156425408147>
113. Rudolph J, Tan S, Tan S (2023) ChatGPT: Bullshit spewer or the end of traditional assessments in higher education? *J Appl Learn Teach* 6(1):342–363. <https://doi.org/10.37074/jalt.2023.6.1.9>
114. Scherer R, Siddiq F, Tondeur J (2019) The technology acceptance model (TAM): A meta-analytic structural equation modeling approach to explaining teachers' adoption of digital technology in education. *Comput Educ* 128:13–35. <https://doi.org/10.1016/j.compedu.2018.09.009>
115. Selwyn N (2023) Should robots replace teachers? AI and the future of education. *Polity*
116. Shaw BP (2024) Artificial intelligence and assessment: Three implications for music educators. *Music Educators J* 111(2):45–52. <https://doi.org/10.1177/00274321241296118>
117. Slavin RE (1986) Best-evidence synthesis: An alternative to meta-analytic and traditional reviews. *Educational Researcher* 15(9):5–11. <https://doi.org/10.3102/0013189X015009005>
118. Song L (2024) Design and implementation of remote piano teaching based on attention-induced multi-head convolutional neural network optimized with hunter-prey optimization. *Int J Comput Intell Syst* 17., Article 379. <https://doi.org/10.1007/s44196-023-00379-3>
119. Spahn C, Voltmer E, Mornell A, Nusseck M (2023) Health-promoting strategies and self-regulated practice behavior in music students. *Front Psychol* 14., Article 1130610. <https://doi.org/10.3389/fpsyg.2023.1130610>
120. Steenbergen-Hu S, Cooper H (2014) A meta-analysis of the effectiveness of intelligent tutoring systems on college students' academic learning. *J Educ Psychol* 106(2):331–347. <https://doi.org/10.1037/a0034752>
121. Strielkowski W, Grebennikova V, Lisovskiy A, Rakhimova G, Vasileva T (2025) AI-driven adaptive learning for sustainable educational transformation. *Sustain Dev* 33(2):1921–1947. <https://doi.org/10.1002/sd.3190>
122. Su J, Zhong Y (2024) Artificial intelligence in teaching and teacher professional development: A systematic review. *Computers Education: Artif Intell* 6., Article 100212. <https://doi.org/10.1016/j.caeai.2024.100212>
123. Tang J (2025) The role of artificial intelligence in improving the efficiency of piano online learning. *J Comput Methods Sci Eng* 25(2):1504–1518. <https://doi.org/10.1177/14727978241300381>

124. Tondeur J, van Braak J, Ertmer PA, Ottenbreit-Leftwich A (2017) Understanding the relationship between teachers' pedagogical beliefs and technology use in education: A systematic review of qualitative evidence. *Education Tech Research Dev* 65(3):555–575. <https://doi.org/10.1007/s11423-016-9481-2>
125. Tuomi I (2024) In search of artificial intelligence (AI) literacy in teacher education: A scoping review. *Computers Education: Artif Intell* 6., Article 100200. <https://doi.org/10.1016/j.caeai.2024.100200>
126. UNESCO, Organization C (2021) Recommendation on the ethics of artificial intelligence. United Nations Educational. <https://www.unesco.org/en/articles/recommendation-ethics-artificial-intelligence>
127. UNESCO (2023) *Guidance for generative AI in education and research*. United Nations Educational, Scientific and Cultural Organization. <https://unesdoc.unesco.org/ark:/48223/pf0000386693>
128. UNESCO (2024) *AI competency framework for teachers*. United Nations Educational, Scientific and Cultural Organization. <https://unesdoc.unesco.org/ark:/48223/pf0000391104>
129. VanLehn K (2011) The relative effectiveness of human tutoring, intelligent tutoring systems, and other tutoring systems. *Educational Psychol* 46(4):197–221. <https://doi.org/10.1080/00461520.2011.611369>
130. Venkatesh V, Thong JY, Xu X (2012) Consumer acceptance and use of information technology: Extending the unified theory of acceptance and use of technology. *MIS Q* 36(1):157–178. <https://doi.org/10.2307/41410412>
131. Wang H, Tlili A (2024) Does ChatGPT enhance student learning? A systematic review and meta-analysis of experimental studies. *Comput Educ* 223., Article 105148. <https://doi.org/10.1016/j.compedu.2024.105148>
132. Wang S (2025) Hybrid models of piano instruction: How combining traditional teaching methods with personalized AI feedback affects learners' skill acquisition, self-efficacy, and academic locus of control. *Educ Inform Technol* 30(2):1245–1268. <https://doi.org/10.1007/s10639-025-13359-1>
133. Wang YH (2023) Exploring the effects of using various designs of game-based materials on music learning. *Interact Learn Environ* 31(5):2650–2664. <https://doi.org/10.1080/10494820.2021.1916768>
134. Woolf BP (2010) Building intelligent interactive tutors: Student-centered strategies for revolutionizing e-learning. Morgan Kaufmann. <https://doi.org/10.1016/B978-0-12-373594-2.X0001-6>
135. Xiao X (2024) Innovative application of artificial intelligence in piano music education and evaluation. *J Comput Methods Sci Eng* 24(3):1473–1486. <https://doi.org/10.3233/JCM-247311>
136. Xu W, Ouyang F (2022) The application of AI technologies in STEM education: A systematic review from 2011 to 2021. *Int J STEM Educ* 9(1) Article 59. <https://doi.org/10.1186/s40594-022-00377-5>
137. Yang SJ, Ogata H, Matsui T, Chen NS (2021) Human-centered artificial intelligence in education: Seeing the invisible through the visible. *Computers and Education: Artificial Intelligence*, 2, Article 100008. <https://doi.org/10.1016/j.caeai.2021.100008>
138. Yang W, Shen L, Huang CF, Lee J, Zhao X (2024) Development status, frontier hotspots, and technical evaluations in the field of AI music composition since the 21st century: A systematic

- review. IEEE Access 12:89452–89466. <https://doi.org/10.1109/ACCESS.2024.3419789>
139. Yang Y, Welch G (2023) A systematic literature review of Chinese music education studies during 2007 to 2019. *Int J Music Educ* 41(2):175–198. <https://doi.org/10.1177/02557614221094822>
 140. Yilmaz R, Yilmaz FGK (2023) The effect of generative artificial intelligence (AI)-based tool use on students' computational thinking skills, programming self-efficacy and motivation. *Computers Education: Artif Intell* 4:100147. <https://doi.org/10.1016/j.caeai.2023.100147>
 141. Yin Y, Dong Y (2025) An artificial intelligence-based interactive learning environment for music education in China: Traditional Chinese music and its contemporary development as a way to increase cultural capital. *Eur J Educ* 60(1) Article e12858. <https://doi.org/10.1111/ejed.12858>
 142. Yu J (2021) Automatic evaluation system for piano performance based on the internet of things technology under the background of artificial intelligence. *Math Probl Eng* 2021(1) Article 8501960. <https://doi.org/10.1155/2021/8501960>
 143. Zawacki-Richter O, Marín VI, Bond M, Gouverneur F (2019) Systematic review of research on artificial intelligence applications in higher education. *Int J Educational Technol High Educ* 16., Article 39. <https://doi.org/10.1186/s41239-019-0171-0>
 144. Zelenak MS (2024) Self-efficacy and music performance: A meta-analysis. *Psychol Music* 52(2):147–162. <https://doi.org/10.1177/03057356231222432>
 145. Zhai C, Wibowo S, Li LD (2024) The effects of over-reliance on AI dialogue systems on students' cognitive abilities: A systematic review. *Smart Learn Environ* 11(1). Article 28. <https://doi.org/10.1186/s40561-024-00310-9>
 146. Zhai X, Xu J (2022) AI-supported learning and assessment: A review of emerging educational applications. *Comput Educ* 3., Article 100072. <https://doi.org/10.1016/j.caeai.2022.100072>
 147. Zhang K, Aslan AB (2021) AI technologies for education: Recent research & future directions. *Computers Education: Artif Intell* 2., Article 100025. <https://doi.org/10.1016/j.caeai.2021.100025>
 148. Zhang M (2025) Advancing deep learning for expressive music composition and performance modeling. *Sci Rep* 15(1). Article 28007. <https://doi.org/10.1038/s41598-025-12345-6>
 149. Zhang Q (2025) *Integrating piano learning software with traditional piano teaching* [Doctoral dissertation, University of Miami]. ProQuest Dissertations Publishing
 150. Zhao X, Guo Z, Liu S (2021) Exploring key competencies and professional development of music teachers in primary schools in the era of artificial intelligence. *Sci Program* 2021(1):5097003. <https://doi.org/10.1155/2021/5097003>
 151. Zhao X, Liu Y, Chen Z (2023) Audio–visual fusion-based piano performance assessment using deep learning. *Appl Sci* 13(5) Article 2890. <https://doi.org/10.3390/app13052890>
 152. Zhou Z, Wu Y, Wu Z, Zhang X, Yuan R, Ma Y, Zhu J, Guo Y (2024) Can LLMs reason in music? An evaluation of LLMs' capability of music understanding and generation. *arXiv preprint arXiv* 240721531. <https://doi.org/10.48550/arXiv.2407.21531>

153. Zi H (2023) Intelligent piano teaching system and method based on cloud platform. In *2023 International conference on distributed computing and electrical circuits and electronics (ICDCECE)* (pp. 1–7). IEEE. <https://doi.org/10.1109/ICDCECE57866.2023.10150890>

154. Zimmerman BJ (2002) Becoming a self-regulated learner: An overview. *Theory into Pract* 41(2):64–70. https://doi.org/10.1207/s15430421tip4102_2

Figures

PRISMA 2020 flow diagram for new systematic reviews which included searches of databases and registers only

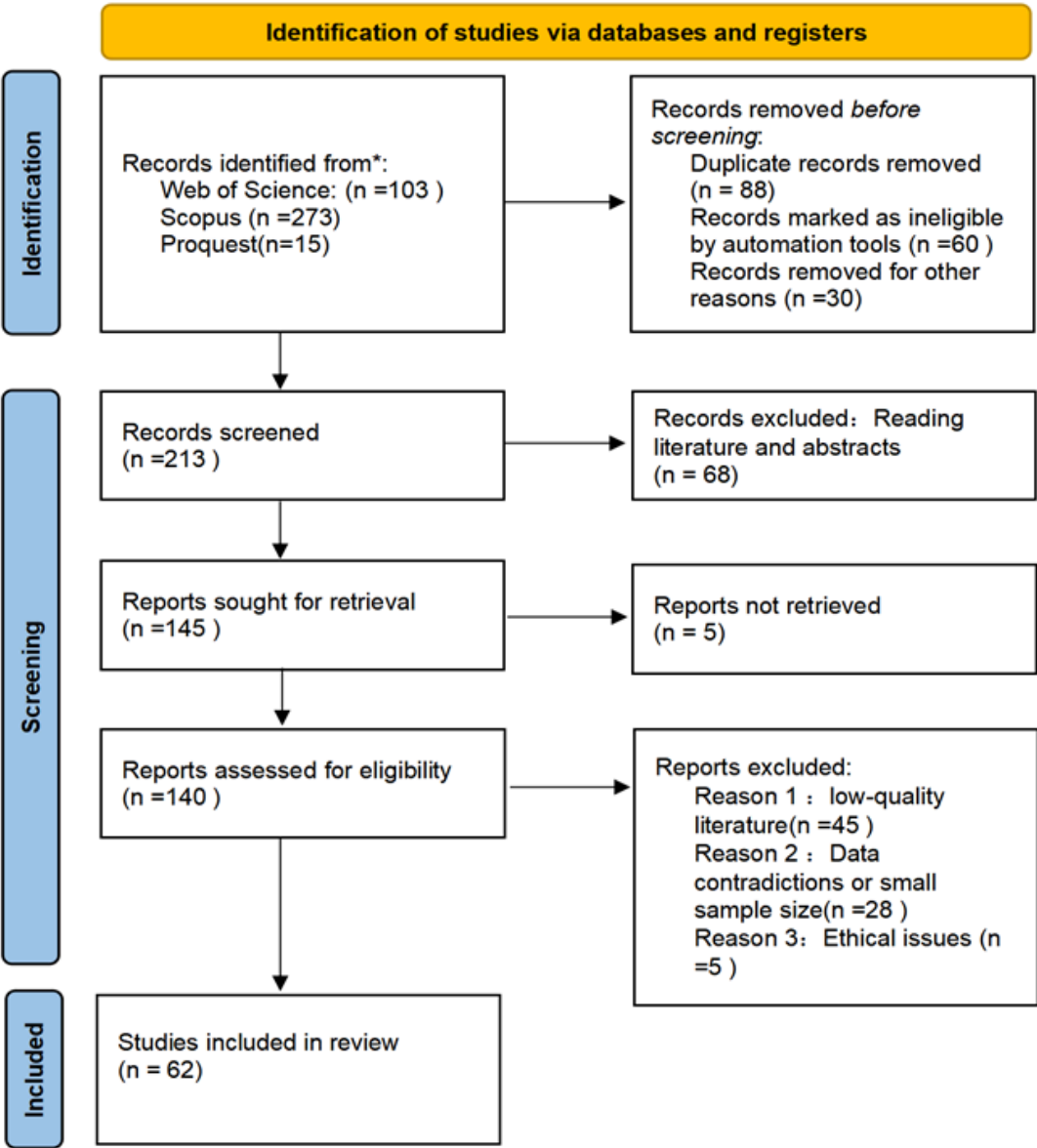


Figure 1

PRISMA Flow Diagram of Database Searching Process.

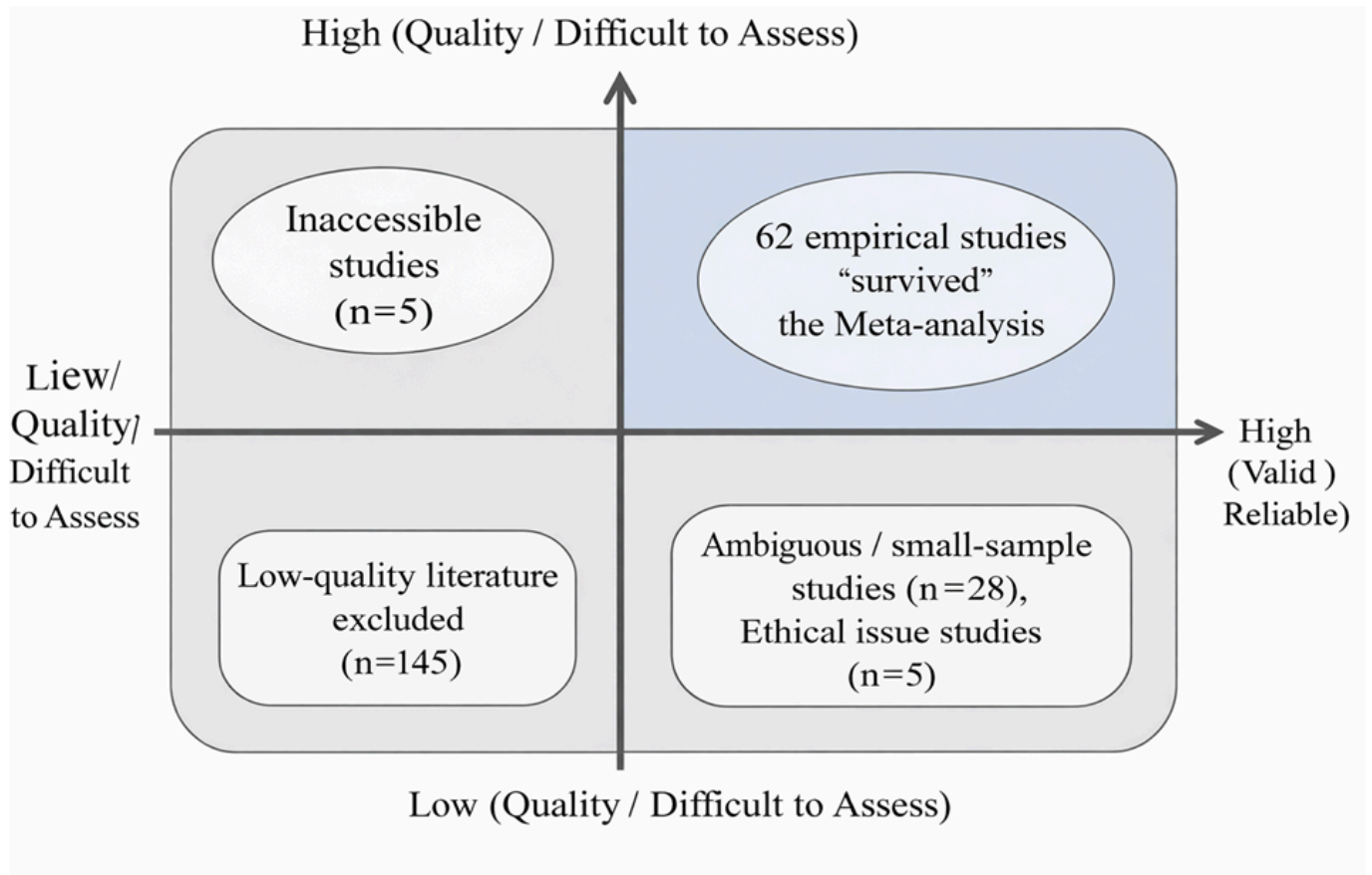


Figure 2

Cartesian Coordinate System for Quality Assessment

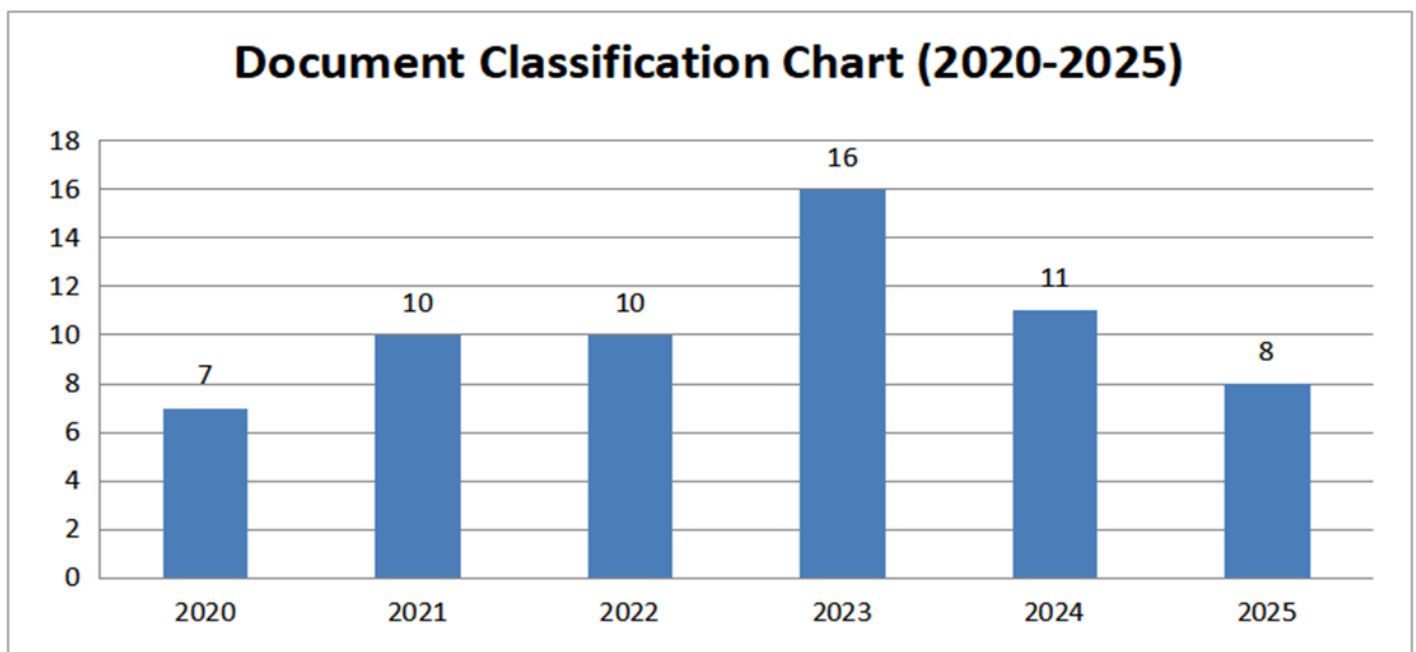


Figure 3

Temporal Distribution of Publications (2020–2025)

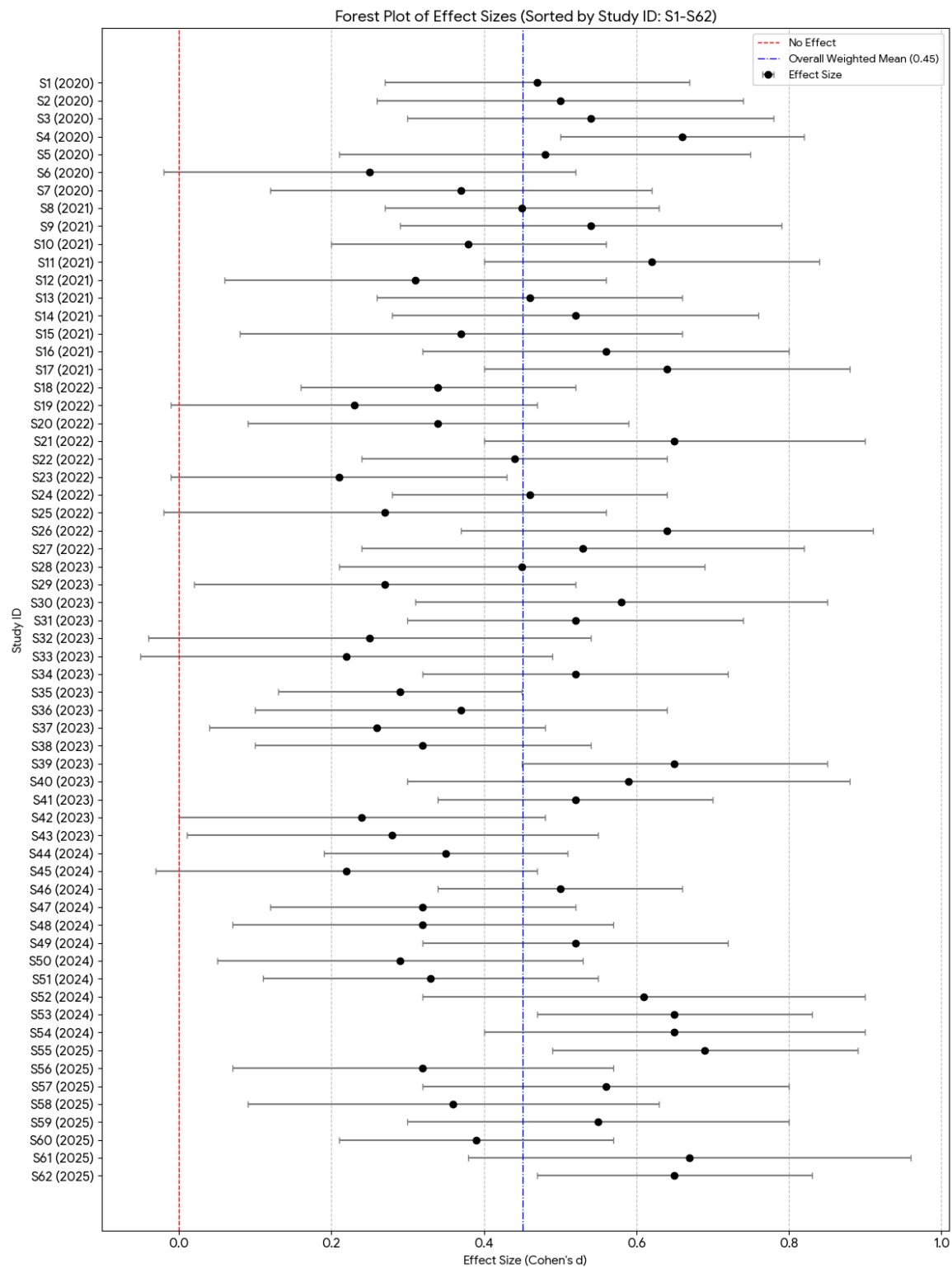
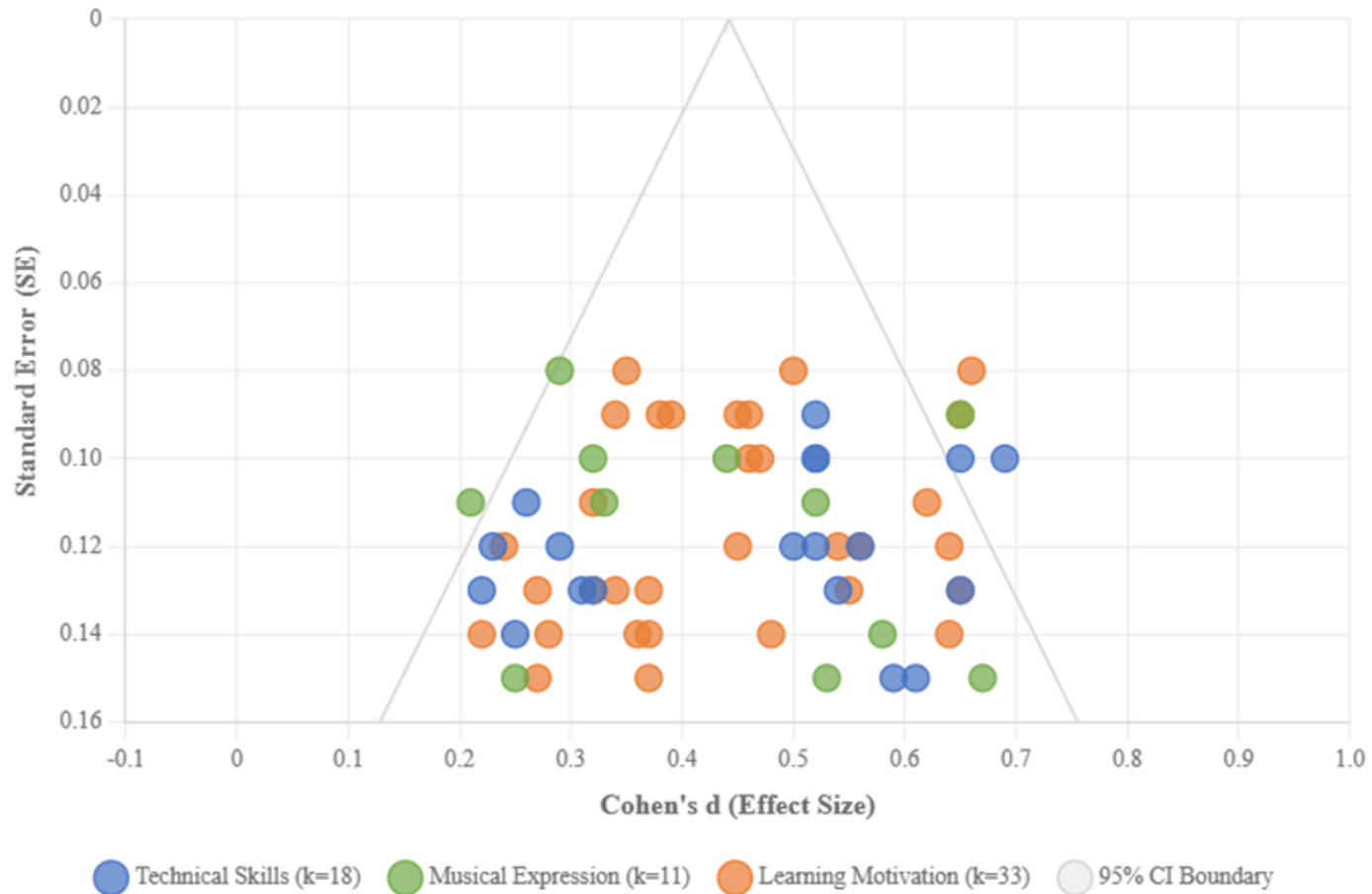


Figure 4

Forest Plot of Effect Sizes for AI-Supported Piano Education Interventions (2020–2025)

Note. This forest plot displays standardized effect sizes (Cohen's d) and 95% confidence intervals for all 62 included studies, organized chronologically by publication year. Each horizontal line represents a

single study's effect size estimate and confidence interval. The vertical dashed line at $d = 0$ indicates no effect. Studies with confidence intervals not crossing the zero line demonstrate statistically significant effects. The diamond marker at the bottom represents the overall pooled effect size ($d = 0.442$, 95% CI [0.398, 0.486]) calculated using random-effects meta-analysis. Color coding indicates instructional domain: blue = Technical Skills ($k = 18$), green = Musical Expression ($k = 11$), orange = Learning Motivation ($k = 33$). The width of confidence intervals reflects sample size and measurement precision, with narrower intervals indicating more precise estimates. Visual inspection reveals predominantly positive effects across all domains and years, with substantial between-study variability ($I^2 = 61.1\%$) supporting the use of random-effects modeling.



Publication Bias Assessment:

- Egger's Regression Test: $z = 1.31$, $p = .191$ (not significant)
- Pooled Effect Size: $d = 0.442$ (95% CI [0.398, 0.486])
- Number of Studies: $k = 62$

Figure 5

Funnel Plot for Assessment of Publication Bias

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- [APPENDIX.docx](#)